Fusion Voting

A Unifying Treatise across Combinatorics, Political Science, Physics, Astronomy, and Machine Learning

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Abstract

Fusion voting—the electoral mechanism by which a single candidate may appear on the ballot under the simultaneous endorsement of two or more distinct political parties—is formalised here as a problem of maximum cardinality in a set-valued function on a totally ordered party system. An optimal linear-time algorithm is proved correct via loop-invariant induction and tight in the comparison model; its execution is illustrated with a numerical trace. A score generating function connects the polynomial $G(z) = \sum_j z^{|M_j|}$ to the ℓ^1 norm of the endorsement matrix Σ and yields mean and variance identities.

Seventeen formal results (two theorems, one corollary, three lemmas, ten propositions, one definition) develop rigorous analogies across seven disciplines. *Statistical mechanics:* the full Ising Hamiltonian with ferromagnetic and antiferromagnetic couplings; a theorem proving the ferromagnetic ground state coincides with the cardinality winner; a renormalisation-group party-merger lemma with monotone winner-set flow. *Astrophysics:* multi-spectral source classification; Hertzsprung–Russell stellar phase zones; gravitational lensing as repeated cross-endorsement; the Chandrasekhar limit as an electoral stability threshold; photometric redshift as a band-detection classifier. *Machine learning:* multi-label classification; PAC learning with rank-r sample-complexity bounds; Bayesian Beta-conjugate inference; attention mechanisms as a soft endorsement head; graph neural networks on the bipartite endorsement graph. *Data science:* set-cover NP-hardness; SVD ideological biplot with a spectral cardinality bound; nuclear-norm matrix completion with the Candès–Recht recovery guarantee; Markov chain endorsement dynamics; information-theoretic BSC channel capacity with a Fano lower bound; topological data analysis with Betti-number bounds on the endorsement complex. *Quantum computing:* QUBO formulation; quantum annealing via adiabatic evolution of the Ising Hamiltonian. *Network science:* bipartite endorsement graph; spectral gap; modularity; DeGroot opinion dynamics with a consensus-ordering proposition; SIS epidemic contagion model with an endemic-equilibrium proposition.

Strategic equilibria are established via a potential-game proposition; Arrow's impossibility theorem is audited; a stability proposition characterises the winning margin Δ. International comparative context spans New York, Chile, Belgium, and Germany. The exposition is accompanied by twelve TikZ figures, ten `booktabs` tables, and a structured glossary of forty-three terms.

The treatise ends with "The End"

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1 Introduction

Electoral systems encode the preferences and coalitions of a polity into a discrete outcome. Among the many mechanisms that have been devised, *fusion voting* (also called *cross-endorsement voting*) occupies a singular position: it allows a single candidate to be simultaneously nominated by multiple parties, with votes cast on any party line summing toward that candidate's total tally.

The combinatorial skeleton of fusion voting is deceptively simple—it is a problem of finding maxima over set cardinalities—yet its ramifications span political equilibria, coalition dynamics, social welfare functions, and, as we shall demonstrate, formal analogues in astrophysics and statistical learning. The present treatise aims to make these connections explicit and rigorous.

1.1 Historical Background

Fusion voting was commonplace throughout the nineteenth-century United States. The Populist Party's cross-endorsement of Democratic presidential nominee William Jennings Bryan in 1896 is perhaps the most celebrated instance [10]. Legislation curtailing the practice—most notably Minnesota's 1901 anti-fusion law, later upheld in *Timmons v. Twin Cities Area New Party* (1997) [2]—gradually restricted its use across most states. New York remains the most prominent jurisdiction where fusion voting is actively practised today; the Working Families Party and the Conservative Party routinely cross-endorse major-party candidates, and their ballot lines can decide close races.

1.2 Scope and Organisation

Section 2 introduces the combinatorial model including monotonicity, generating function, and weighted-affiliation variant. Section 3 proves correctness, complexity, and optimality of the maximum-cardinality algorithm. Section 4 develops the full Ising Hamiltonian analogy including coupling graphs, a ferromagnetic ground-state theorem, and a renormalisation-group party-merger lemma. Section 5 covers multi-spectral classification, the HR diagram, gravitational lensing, the Chandrasekhar stability threshold, and photometric redshift. Section 6 treats multi-label classification, ensemble methods, PAC learning, Bayesian inference, attention mechanisms, and graph neural networks. Section 7 covers the endorsement matrix, set-cover, SVD with spectral bound, matrix completion, information-theoretic capacity, and topological data analysis. Section 8 consolidates ten comparative and summary tables. Section 9 presents twelve TikZ figures. Section 10 analyses Nash equilibria, Arrow's theorem, comparative statics, fuzzy generalisations, and six open problems. Section 11 develops quantum annealing, network science, DeGroot dynamics, the SIS epidemic model, and international comparisons. Section 12 concludes. The glossary and bibliography follow.

2 Formal Combinatorial Model

2.1 Party System as a Totally Ordered Sequence

Definition 2.1 (Party System)**.** *A* party system *is a finite totally ordered sequence*

$$\mathcal{P} = (P_1, P_2, \ldots, P_p), \quad p \in \mathbb{N},$$

where each party P_i has a non-negative integer membership size $n(P_i) \in \mathbb{N}_0$*, and the ordering satisfies*

$$n(P_i) \;\geq\; n(P_{i+1}), \qquad i = 1, 2, \ldots, p-1.$$

Remark 2.2. *The condition $n(P_i) \geq n(P_{i+1})$ is a non-strict inequality; parties of equal size may appear in any relative order without affecting subsequent results.*

The ordering captures relative party size and plays no direct role in the cardinality maximisation problem of Section 3, but becomes essential in the weighted-affiliation variant of Section 2.4.

2.2 Candidate Space and Membership Sets

Definition 2.3 (Candidate and Membership Set)**.** *Let $\mathcal{C} = \{C_1, C_2, \ldots, C_c\}$, with $c \geq 2$, be a finite set of* candidates*. To each candidate C_j we associate a non-empty* membership set

$$M_j \;\subseteq\; \mathcal{P}, \qquad 1 \;\leq\; |M_j| \;\leq\; p.$$

The collection $\mathcal{M} = \{M_1, \ldots, M_c\}$ is the affiliation profile *of the election.*

Remark 2.4. *The power set $2^{\mathcal{P}}$ has cardinality 2^p. Excluding the empty set (every candidate belongs to at least one party), the affiliation profile lives in the product space $(2^{\mathcal{P}} \setminus \{\varnothing\})^c$, which has $(2^p - 1)^c$ elements.*

2.3 The Maximum-Cardinality Problem

Definition 2.5 (Winner Set)**.** *The* winner set *under fusion-cardinality scoring is*

$$\mathcal{W} \;=\; \operatorname{argmax}_{C_j \in \mathcal{C}} |M_j| \,.$$

Remark 2.6. *$\mathcal{W}$ may contain more than one candidate; ties are structurally possible whenever two or more candidates share the maximum cardinality, though they need not share the same parties.*

Lemma 2.7 (Monotonicity of Cardinality Scoring)**.** *For any two candidates $C_i, C_j \in \mathcal{C}$, if $M_i \subseteq M_j$ then $|M_i| \leq |M_j|$, and hence $C_i \in \mathcal{W}$ implies $C_j \in \mathcal{W}$.*

Proof. The first claim is immediate: subset inclusion implies no greater cardinality. For the second, if $C_i \in \mathcal{W}$ then $|M_i| = \max_k |M_k|$. Since $M_i \subseteq M_j$ gives $|M_j| \geq |M_i|$, we have $|M_j| = \max_k |M_k|$ as well, so $C_j \in \mathcal{W}$. □

Remark 2.8. *Lemma 2.7 shows that cardinality scoring is* monotone *with respect to the partial order $(2^{\mathcal{P}}, \subseteq)$: moving a candidate "up" in the inclusion lattice never decreases their score. This monotonicity is* not *shared by all social choice rules; the Borda count, for instance, can rank a candidate lower when a new alternative is removed from the ballot (independence-of-irrelevant-alternatives failure).*

2.4 Weighted Affiliation Variant

When party sizes matter, one replaces the plain cardinality score with the *weighted affiliation score*:

$$w(C_j) \;=\; \sum_{P_i \in M_j} n(P_i).$$

The winner set becomes $\mathcal{W}_w = \operatorname{argmax}_{C_j \in \mathcal{C}} w(C_j)$. All algorithmic results in Section 3 apply to this variant by substituting $w(C_j)$ for $|M_j|$ as the comparison key.

2.5 Generating Function of the Score Distribution

The distribution of fusion-cardinality scores across all candidates is summarised by the *score generating function*

$$G(z) \;=\; \sum_{j=1}^{c} z^{|M_j|}, \qquad z \in \mathbb{C}.$$

Lemma 2.9 (Generating-Function Identities)**.** *The following identities hold:*

(i) $G(1) = c$ *(total number of candidates).*

(ii) $G'(1) = \sum_{j=1}^{c} |M_j| = \|\Sigma\|_1$ *(sum of all fusion-cardinality scores, equal to the ℓ^1 norm of Σ).*

(iii) $\bar{k} := G'(1)/c$ *is the mean fusion-cardinality score.*

(iv) The variance of the scores is $\mathrm{Var}(k) = G''(1)/c + \bar{k}(1-\bar{k}) - \bar{k}^2 + \bar{k} = G''(1)/c + \bar{k} - \bar{k}^2$.

Proof. (i) Setting $z = 1$ gives $G(1) = \sum_j 1 = c$. (ii) Differentiating term-by-term, $G'(z) = \sum_j |M_j|\, z^{|M_j|-1}$, so $G'(1) = \sum_j |M_j|$. Since $\Sigma_{ji} = \sigma_i^{(j)}$, we have $\sum_j |M_j| = \sum_j \sum_i \Sigma_{ji} = \|\Sigma\|_1$. (iii) follows immediately. (iv) $G''(1) = \sum_j |M_j|\,(|M_j| - 1)$, so $G''(1)/c = \mathbb{E}[k^2] - \mathbb{E}[k]$; substituting $\mathrm{Var}(k) = \mathbb{E}[k^2] - (\mathbb{E}[k])^2$ yields the stated expression. □

Remark 2.10. *The generating function $G(z)$ is a polynomial of degree at most p with non-negative integer coefficients summing to c. Its roots lie in $\mathbb{C}$ and encode the full score distribution. In the degenerate case where all candidates have equal score k_0, $G(z) = c\, z^{k_0}$—a monomial. In the maximally dispersed case where scores are distinct integers $1, 2, \ldots, c$, $G(z) = \sum_{j=1}^{c} z^j = z(z^c - 1)/(z - 1)$, whose roots are the c-th roots of unity scaled by z.*

3 Algorithm and Complexity

3.1 The Linear-Scan Procedure

Algorithm 1 identifies $\mathcal{W}$ in a single pass over the candidate list, maintaining a running maximum and a set of current winners.

Algorithm 1 MaxCardWinner($\mathcal{C}, \mathcal{M}$)

Require: Candidate list $C_1, \ldots, C_c$; membership sets $M_1, \ldots, M_c$
Ensure: Winner set $\mathcal{W}$ and maximum cardinality k^*

```
1:  k* ← |M_1|
2:  W ← {C_1}
3:  for j ← 2 to c do
4:      k ← |M_j|
5:      if k > k* then
6:          k* ← k
7:          W ← {C_j}
8:      else if k = k* then
9:          W ← W ∪ {C_j}
10:     end if
11: end for
12: return (W, k*)
```

3.2 Correctness Proof

Theorem 3.1 (Correctness of Algorithm 1)**.** *Upon termination, Algorithm 1 returns* $\mathcal{W} = \operatorname{argmax}_{C_j \in \mathcal{C}} |M_j|$.

Proof. We establish the following loop invariant by induction on j:

Invariant $I(j)$**:** After processing candidate C_j,

$$k^* = \max_{1 \le \ell \le j} |M_\ell| \quad \text{and} \quad \mathcal{W} = \left\{C_\ell : \ell \le j,\ |M_\ell| = k^*\right\}.$$

Base case ($j = 1$). Lines 1–2 set $k^* = |M_1|$ and $\mathcal{W} = \{C_1\}$. The maximum over the singleton $\{C_1\}$ is indeed $|M_1|$, so $I(1)$ holds.

Inductive step. Assume $I(j-1)$ holds for some $j \ge 2$ and let $k = |M_j|$. Three mutually exclusive cases arise:

- *Case* $k > k^*$. Then $k = \max_{1 \le \ell \le j} |M_\ell|$. Lines 6–7 update $k^* \leftarrow k$ and $\mathcal{W} \leftarrow \{C_j\}$, re-establishing $I(j)$.
- *Case* $k = k^*$. The maximum is unchanged. Line 9 sets $\mathcal{W} \leftarrow \mathcal{W} \cup \{C_j\}$, adding the new co-winner. $I(j)$ holds.
- *Case* $k < k^*$. Neither k^* nor $\mathcal{W}$ changes. Since $k < k^*$, $C_j \notin \mathcal{W}$. $I(j)$ holds trivially.

By induction, $I(c)$ holds at termination, which is exactly the statement of the theorem. □

3.3 Complexity Analysis

Theorem 3.2 (Time and Space Complexity)**.** *Algorithm 1 runs in* $\Theta(c)$ *time and* $O(c)$ *space, assuming membership sizes are precomputed. If not precomputed, the time rises to* $O(cp)$.

Proof. The loop body executes exactly $c - 1$ times. Each iteration performs a single integer comparison and, in at most one branch, a set-union costing $O(1)$ (pointer insertion). Initialisation is $O(1)$. Hence the total time is $\Theta(c)$. The winner set $\mathcal{W}$ stores at most c references, giving $O(c)$ auxiliary space. If $|M_j|$ must be computed on the fly, iterating over M_j costs $O(p)$ per candidate, raising the total to $O(cp)$. □

Corollary 3.3 (Optimality in the Comparison Model)**.** *No comparison-based algorithm can identify* $\mathcal{W}$ *in fewer than* $c - 1$ *comparisons in the worst case.*

Proof. With fewer than $c - 1$ comparisons, at least one candidate remains unexamined. An adversary may then assign that candidate any cardinality, potentially contradicting the algorithm's output. □

4 Statistical Mechanics Analogy

The structure of fusion voting bears a precise resemblance to the *Ising model* of ferromagnetism [4] and, more generally, to *spin-glass* systems [5].

4.1 Spin-Lattice Correspondence

Consider a one-dimensional lattice $\Lambda = \{1, 2, \ldots, p\}$ where each site i corresponds to party P_i. For candidate C_j, define a *spin configuration* $\boldsymbol{\sigma}^{(j)} \in \{0,1\}^p$ by

$$\sigma_i^{(j)} \;=\; \begin{cases} 1 & \text{if } P_i \in M_j, \\ 0 & \text{otherwise.} \end{cases}$$

Then $|M_j| = \left\|\boldsymbol{\sigma}^{(j)}\right\|_1 = \sum_{i=1}^{p} \sigma_i^{(j)}$, the *total magnetisation* of the configuration. The maximum-cardinality problem becomes:

$$\mathcal{W} \;=\; \operatorname{argmax}_j \; \left\|\boldsymbol{\sigma}^{(j)}\right\|_1,$$

the candidate with the highest magnetisation.

4.2 Hamiltonian Formulation

Define an external field $h_i = n(P_i)$ (party membership size as field strength) and the *party field vector* $\boldsymbol{h} = (n(P_1), \ldots, n(P_p))^\top$. The weighted affiliation score of Section 2.4 is then a *Zeeman coupling*:

$$w(C_j) \;=\; \boldsymbol{h} \cdot \boldsymbol{\sigma}^{(j)} \;=\; \sum_{i=1}^{p} h_i \, \sigma_i^{(j)}.$$

Finding the maximum-weighted candidate is equivalent to finding the spin configuration of *minimum Zeeman energy* aligned with the external field—a zero-temperature ground-state problem.

4.3 Partition Function and Electoral Entropy

At finite (metaphorical) temperature $T > 0$, the probability that a randomly sampled voter perceives candidate C_j as most-endorsed is given by the *Boltzmann* (softmax) distribution:

$$\mathbb{P}(C_j \text{ wins}) \;=\; \frac{\exp\big(|M_j|\,/T\big)}{\displaystyle\sum_{k=1}^{c} \exp\big(|M_k|\,/T\big)}.$$

The *Shannon entropy* [14] of this distribution,

$$H(T) \;=\; -\sum_{j=1}^{c} \mathbb{P}(C_j) \, \log \mathbb{P}(C_j),$$

measures *electoral uncertainty*. As $T \to 0^+$, $H \to 0$ and the distribution concentrates on $\mathcal{W}$ (the ground state). As $T \to \infty$, $H \to \log c$, corresponding to a uniform distribution and maximum uncertainty.

4.4 Inter-Party Interactions and the Full Ising Hamiltonian

The model above treats parties as independent sites (no coupling between them). A richer analogy admits *pairwise interactions* between parties, capturing political alignment or opposition. Define coupling constants $J_{ij} \in \mathbb{R}$ for each pair (P_i, P_j), where $J_{ij} > 0$ indicates that the two parties tend to endorse the same candidate (ferromagnetic, allied parties) and $J_{ij} < 0$ indicates they tend to endorse opposing candidates (antiferromagnetic, rival parties). The full *Ising Hamiltonian* for candidate C_k's configuration $\boldsymbol{\sigma}^{(k)}$ is then

$$\mathcal{H}\Big(\boldsymbol{\sigma}^{(k)}\Big) \;=\; -\sum_{1 \le i < j \le p} J_{ij} \, \sigma_i^{(k)} \sigma_j^{(k)} \;-\; \sum_{i=1}^{p} h_i \, \sigma_i^{(k)},$$

where the first sum runs over all party pairs and the second is the Zeeman term from Section 4. The ground-state candidate under $\mathcal{H}$ is not necessarily the fusion-cardinality winner; the coupling term penalises configurations where two rival parties ($J_{ij} < 0$) both endorse the same candidate, reflecting the electoral reality that cross-endorsements between ideological opposites can be strategically counterproductive.

Figure 5 illustrates a four-party interaction graph for the running example, with coupling signs annotating the edges.

4.5 Renormalisation Group and Party Coarse-Graining

The *renormalisation group* (RG) is a framework in statistical mechanics for systematically integrating out short-scale degrees of freedom to obtain an effective theory at a coarser scale [12]. In the electoral analogy, an *RG step* corresponds to *merging* two parties P_i and $P_{i'}$ into a single effective party $\tilde{P}_{ii'}$, reducing the party system from p to $p-1$ elements.

Definition 4.1 (Party Merger)**.** *Given a party system $\mathcal{P} = (P_1, \ldots, P_p)$ and a pair $\{i, i'\}$, the* merged party system *$\tilde{\mathcal{P}}$ replaces P_i and $P_{i'}$ with $\tilde{P}_{ii'}$, whose membership size is $n(\tilde{P}_{ii'}) = n(P_i) + n(P_{i'})$, and whose endorsement of candidate C_j is*

$$\tilde{\sigma}_{ii',j} \;=\; \max(\sigma_{i,j},\, \sigma_{i',j}) \;=\; \mathbf{1}[P_i \in M_j \text{ or } P_{i'} \in M_j].$$

Under a party merger, the cardinality score of candidate C_j decreases by exactly one if $P_i \in M_j$ and $P_{i'} \in M_j$ (both parties endorsed C_j, but they now count as one), and is unchanged otherwise.

Lemma 4.2 (Cardinality Under Merger)**.** *Let $\left|\tilde{M}_j\right|$ denote the cardinality score of C_j in the merged party system. Then*

$$\left|\tilde{M}_j\right| \;=\; |M_j| - \mathbf{1}[P_i \in M_j \text{ and } P_{i'} \in M_j].$$

In particular, $\left|\tilde{M}_j\right| \le |M_j|$ always, and the winner set can only shrink or stay the same after a merger: $\mathcal{W}(\tilde{\mathcal{P}}) \supseteq \mathcal{W}(\mathcal{P})$ or $\mathcal{W}(\tilde{\mathcal{P}}) = \mathcal{W}(\mathcal{P})$.

Proof. The merged endorsement $\tilde{\sigma}_{ii',j} = 1$ iff at least one of $P_i, P_{i'}$ endorses C_j. Therefore $\left|\tilde{M}_j\right| = |M_j| - \sigma_{i,j}\sigma_{i',j}$ (subtracting one if both endorsed). Since $\sigma_{i,j}\sigma_{i',j} \in \{0,1\}$, the inequality follows. The winner-set monotonicity follows: if $C_j \in \mathcal{W}(\mathcal{P})$ and the merger reduces $|M_j|$ by 1 while reducing $|M_{j'}|$ for all other $C_{j'} \notin \mathcal{W}(\mathcal{P})$ by at most 1, then C_j remains a winner unless a previously non-winning candidate is brought to the same new maximum. □

Remark 4.3. *Iterating $p-1$ mergers reduces the party system to a single mega-party $\tilde{P}_{1\cdots p}$ endorsed by every candidate. At this coarsest scale, all candidates have cardinality score 1 and the winner set is $\mathcal{C}$ itself. The RG flow thus moves from a* discriminating *regime (many fine-grained parties, distinct scores) to a* non-discriminating *fixed point (one party, all candidates tied). The sequence of winner sets $\mathcal{W}^{(0)} \supseteq \mathcal{W}^{(1)} \supseteq \cdots \supseteq \mathcal{W}^{(p-1)}$ is non-increasing under successive mergers until the trivial fixed point is reached. This RG perspective is illustrated in Figure 10.*

Theorem 4.4 (Ferromagnetic Ground State Agrees with Cardinality Winner)**.** *If all coupling constants are non-negative ($J_{ij} \ge 0$ for all i, j) and all external fields are equal ($h_i = h > 0$ for all i), then the ground-state configuration of $\mathcal{H}$ is the fully magnetised state $\boldsymbol{\sigma}^* = \mathbf{1}_p$ (all sites up), which corresponds to a candidate endorsed by all p parties—the unique fusion-cardinality maximum when such a candidate exists.*

Proof. Under the stated conditions, $\mathcal{H}(\boldsymbol{\sigma}) = -\sum_{i<j} J_{ij}\sigma_i\sigma_j - h\sum_i \sigma_i$. Both sums are non-negative for $\boldsymbol{\sigma} = \mathbf{1}_p$ and achieve their individual maxima there (since $J_{ij} \ge 0$ and $h > 0$). Thus $\mathcal{H}(\mathbf{1}_p) \le \mathcal{H}(\boldsymbol{\sigma})$ for all $\boldsymbol{\sigma} \in \{0,1\}^p$, making $\mathbf{1}_p$ the global minimiser. A candidate C_j with $\boldsymbol{\sigma}^{(j)} = \mathbf{1}_p$ has $|M_j| = p = \max_k |M_k|$, so $C_j \in \mathcal{W}$. □

Example 4.5 (Numerical Illustration)**.** *Let $p = 4$, $c = 3$, and consider the following affiliation profile:*

$$\boldsymbol{\sigma}^{(1)} = (1,1,0,0)^\top, \quad \boldsymbol{\sigma}^{(2)} = (1,1,1,1)^\top, \quad \boldsymbol{\sigma}^{(3)} = (0,0,1,0)^\top.$$

With uniform field $h_i = 1$ and all $J_{ij} = 1$ (ferromagnetic), the Hamiltonian values are:

$$\begin{aligned}
\mathcal{H}(\boldsymbol{\sigma}^{(1)}) &= -(J_{12}) - 2h = -1 - 2 = -3, \\
\mathcal{H}(\boldsymbol{\sigma}^{(2)}) &= -(J_{12} + J_{13} + J_{14} + J_{23} + J_{24} + J_{34}) - 4h = -6 - 4 = -10, \\
\mathcal{H}(\boldsymbol{\sigma}^{(3)}) &= 0 - h = -1.
\end{aligned}$$

The ground state is $\boldsymbol{\sigma}^{(2)}$ (minimum $\mathcal{H} = -10$), which is also the cardinality winner ($|M_2| = 4$). Algorithm 1 returns $\mathcal{W} = \{C_2\}$, $k^* = 4$, in two comparisons.

5 Astrophysical Analogy

5.1 Multi-Spectral Source Classification

In observational astronomy, a celestial source (star, galaxy, or quasi-stellar object) is detected across multiple electromagnetic bands: radio, infrared, optical, ultraviolet, X-ray, and γ-ray [13]. The correspondence with fusion voting is exact:

$$\begin{aligned}
\mathcal{P} &\longleftrightarrow \text{ spectral bands } \{B_1, \ldots, B_p\}, \\
C_j &\longleftrightarrow \text{ celestial source } s_j, \\
M_j &\longleftrightarrow \text{ set of bands in which } s_j \text{ is detected.}
\end{aligned}$$

The analogue of the fusion-voting winner is the *most multi-band source*—the source detected across the greatest number of spectral windows, a primary figure of merit in multi-messenger astronomy.

5.2 Spectral Energy Distribution Vector

The weighted affiliation score $w(C_j) = \boldsymbol{h} \cdot \boldsymbol{\sigma}^{(j)}$ maps to a *weighted flux sum*:

$$F_j \;=\; \sum_{i=1}^{p} F_{ij}\, \sigma_i^{(j)},$$

where F_{ij} is the flux of source s_j in band B_i and $n(P_i) \leftrightarrow F_i^{\text{ref}}$ is a reference flux (for example, the median catalogue flux in band B_i). Finding the candidate with maximum weighted affiliation is then precisely *integrated broadband luminosity ranking.*

5.3 Hertzsprung–Russell Correspondence

In the Hertzsprung–Russell (HR) diagram, a star's position is determined by its luminosity L and effective temperature T_{eff}. Different evolutionary phases (main sequence, red giant branch, horizontal branch, white dwarf) correspond to distinct *party affiliations.* A star undergoing a transition—for example, a subgiant crossing from the main sequence to the red giant branch—simultaneously satisfies membership criteria for two evolutionary phases, in direct correspondence with a cross-endorsed candidate.

The fraction of its lifetime that a star spends in evolutionary phase ϕ is

$$f_\phi \;=\; \frac{\tau_\phi}{\sum_{\phi'} \tau_{\phi'}},$$

analogous to the probability weight a voter assigns to a candidate's party line when allocating electoral trust.

5.4 Gravitational Lensing as Repeated Cross-Endorsement

A second astrophysical analogy arises in *gravitational lensing.* When a massive foreground object (a galaxy cluster or dark-matter halo) lies between a distant source and the observer, general relativity

bends the source's light into multiple images. Each *image* is an independent detection of the same source at a distinct sky position, in analogy with a single candidate appearing on multiple ballot lines:

$$\begin{aligned} \text{Background source} &\longleftrightarrow \text{Candidate } C_j, \\ \text{Gravitational image } \ell &\longleftrightarrow \text{Party-line endorsement } P_i, \\ \text{Number of images } |M_j| &\longleftrightarrow \text{Fusion-cardinality score.} \end{aligned}$$

A *quad lens* (four images) corresponds to $|M_j| = 4$, while a *double lens* corresponds to $|M_j| = 2$. The time delay between images—the *Shapiro delay*—encodes the lens mass, just as the weighting $n(P_i)$ encodes each party's size. The weighted flux sum $F_j = \sum_i F_{ij}\sigma_i^{(j)}$ is the analogue of the *total magnified flux*, which is maximised for the most strongly lensed source—the candidate endorsed by the most and largest parties.

The total magnification μ_{tot} obeys the relation $\mu_{\text{tot}} = \sum_i |\mu_i|$, where $|\mu_i|$ is the magnification of image i. Under the identification $|\mu_i| \leftrightarrow n(P_i)$, this is precisely $w(C_j)$. Strong-lensing surveys therefore solve an instance of the weighted-affiliation maximisation problem of Section 2.4 at cosmological scale.

5.5 Chandrasekhar Limit and the Winning-Margin Threshold

A white dwarf star is stable as long as its mass M is below the *Chandrasekhar limit* $M_{\text{Ch}} \approx 1.44\, M_\odot$. Beyond this threshold, electron degeneracy pressure cannot resist gravitational collapse and the star transitions to a neutron star or black hole [24]. The Chandrasekhar limit is a *stability threshold*: below it, perturbations to the stellar mass are absorbed; at or above it, a qualitative phase transition occurs.

This is precisely the structure of Proposition 10.3: the *winning margin* Δ is the electoral analogue of the gap $M_{\text{Ch}} - M$. As long as a non-winner's cardinality score is more than $\Delta - 1$ below the maximum, it cannot reach $\mathcal{W}$ by gaining a single endorsement. At $\Delta = 1$—the electoral analogue of a star approaching the Chandrasekhar limit—any single additional endorsement to a runner-up triggers a qualitative change in the winner set.

The analogy extends to the *accretion* mechanism: a binary star system in which a white dwarf accretes mass from a companion (slowly gaining endorsements) until it crosses M_{Ch} and explodes as a Type Ia supernova (the runner-up suddenly joins $\mathcal{W}$). Type Ia supernovae are standard candles precisely because this threshold is universal; the electoral analogue is that the critical margin $\Delta = 1$ is the universal threshold for strategic single-endorsement interventions, regardless of the number of parties or candidates.

Definition 5.1 (Electoral Stability Radius)**.** *The* stability radius *of an election is* $\rho_{\text{stab}} = \Delta - 1 \geq 0$. *At* $\rho_{\text{stab}} = 0$ *the election is* marginally stable*: any endorsement change to any candidate can alter* $\mathcal{W}$. *At* $\rho_{\text{stab}} \geq 1$ *the election is* robustly stable*: at least two coordinated endorsement changes are required to alter the winner set.*

5.6 Photometric Redshift and the Endorsement Classifier

A concrete algorithmic application of the astrophysical analogy arises in *photometric redshift estimation* [13]. Given a galaxy's flux measurements $(F_1, \ldots, F_p)$ across p photometric bands (the counterpart of $\boldsymbol{h} \cdot \boldsymbol{\sigma}^{(j)}$), a classifier must determine the galaxy's redshift z. Treating discrete redshift bins as "candidates" and the detected bands as "parties", Algorithm 1 (applied to band-detection counts) identifies the redshift bin supported by the most photometric bands—a fast surrogate for template-fitting. The endorsement matrix Σ in this context is a binary band-detection matrix of dimension $N_{\text{gal}} \times p$, amenable to all the SVD analysis of Section 7.

6 Machine Learning Connections

6.1 Multi-Label Classification

In machine learning, *multi-label classification* [6] assigns to each instance x a *label set* $L(x) \subseteq \mathcal{L}$, where $\mathcal{L}$ is the label space. The correspondence with fusion voting is immediate:

$$x \longleftrightarrow C_j, \qquad \mathcal{L} \longleftrightarrow \mathcal{P}, \qquad L(x) \longleftrightarrow M_j.$$

The Hamming score of a predicted label set $\hat{L}$ against ground truth L is

$$\text{Hamming}(\hat{L}, L) \;=\; 1 - \frac{1}{p}\left|\hat{L} \triangle L\right|,$$

where $\triangle$ denotes symmetric difference. A perfect predictor of party membership achieves Hamming score 1.

6.2 Ensemble Methods and Fusion Voting

An *ensemble* [7] of p base classifiers $\{h_1, \ldots, h_p\}$ combines individual predictions via some rule. Interpreting party P_i as base classifier h_i and $\sigma_i^{(j)} = 1$ as "h_i endorses C_j", the three canonical combination schemes correspond directly to scoring rules:

- **Plurality (hard) voting.** The winner is the candidate endorsed by the most classifiers, i.e. $\operatorname{argmax}_j |M_j|$—exactly the fusion-cardinality winner.
- **Weighted voting.** Assigning weight $h_i = n(P_i)$ to each classifier yields the weighted affiliation score $w(C_j)$.
- **Stacking.** A meta-learner trained on the endorsement matrix $\Sigma \in \{0,1\}^{c\times p}$ produces a calibrated posterior, analogous to the Boltzmann distribution of Section 4.

6.3 Neural-Network Interpretation

Consider a single-layer linear network with input $\boldsymbol{\sigma}^{(j)} \in \{0,1\}^p$, weight vector $\boldsymbol{w} = \boldsymbol{h}$, and no nonlinear activation:

$$\hat{y}(C_j) \;=\; \boldsymbol{w}^\top \boldsymbol{\sigma}^{(j)} \;=\; w(C_j).$$

The maximum-weighted candidate is the output of an argmax layer applied to $\{\hat{y}(C_j)\}_{j=1}^{c}$. Training this network on historical election outcomes—using cross-entropy loss between predicted winner probabilities and observed results—is a tractable supervised learning problem [11].

6.4 PAC-Learning Bounds for Endorsement Prediction

We may ask: how many historical elections must we observe before a learner can reliably predict the endorsement matrix Σ of a new election? Formulating this in the PAC (Probably Approximately Correct) [15] framework, the hypothesis class $\mathcal{H}$ consists of all binary matrices $\{0,1\}^{c\times p}$, which has VC dimension at most cp. The *fundamental theorem of statistical learning* then guarantees that a sample of

$$m \;\geq\; \frac{1}{\varepsilon}\left(cp\,\ln\frac{e}{\varepsilon} + \ln\frac{1}{\delta}\right)$$

elections suffices to learn a predictor achieving (ε, δ)-PAC accuracy on the Hamming loss. In practice, because the rows of Σ are not independent (parties' endorsement decisions are correlated), tighter bounds follow from structural risk minimisation over a lower-complexity subclass—for example, rank-r Boolean matrices with $r \ll \min(c, p)$, whose VC dimension scales as $O(r(c+p))$ [16].

Proposition 6.1 (Sample Complexity for Endorsement Learning)**.** *Let $\mathcal{H}_r$ denote the class of rank-r Boolean matrices in $\{0,1\}^{c\times p}$. A learner observing*

$$m \;\geq\; \frac{2}{\varepsilon^2}\left(r(c+p)\ln 2 + \ln\frac{2}{\delta}\right)$$

labelled elections drawn i.i.d. from a fixed distribution returns, with probability at least $1-\delta$, a hypothesis $\hat{\Sigma} \in \mathcal{H}_r$ with Hamming loss at most ε above the best achievable in $\mathcal{H}_r$.

Proof. Apply the standard uniform-convergence bound for finite VC classes (Sauer's lemma gives $|\mathcal{H}_r| \leq (em/r(c+p))^{r(c+p)}$), then invoke the Hoeffding–McDiarmid concentration inequality on the bounded Hamming loss function. The stated bound follows from straightforward algebraic simplification. □

6.5 Bayesian Inference over the Endorsement Matrix

Rather than learning a point estimate of Σ, a Bayesian approach maintains a posterior distribution over all possible endorsement matrices. Model each entry Σ_{ji} independently as a Bernoulli random variable with success probability $\theta_{ji} \in (0,1)$, placing an independent Beta prior on each:

$$\theta_{ji} \;\sim\; \mathrm{Beta}(\alpha_{ji},\, \beta_{ji}), \qquad \Sigma_{ji} \mid \theta_{ji} \;\sim\; \mathrm{Bernoulli}(\theta_{ji}).$$

Given n observed elections $\{\Sigma^{(1)}, \ldots, \Sigma^{(n)}\}$, the posterior for each entry is also a Beta distribution by conjugacy:

$$\theta_{ji} \mid \text{data} \;\sim\; \mathrm{Beta}\Big(\alpha_{ji} + \sum_{t=1}^{n} \Sigma_{ji}^{(t)},\; \beta_{ji} + n - \sum_{t=1}^{n} \Sigma_{ji}^{(t)}\Big).$$

The posterior mean $\hat{\theta}_{ji} = (\alpha_{ji} + \sum_t \Sigma_{ji}^{(t)})/(\alpha_{ji} + \beta_{ji} + n)$ provides a Laplace-smoothed estimate of the probability that party P_i endorses candidate C_j in the next election. The expected fusion-cardinality score under the posterior is then

$$\mathbb{E}[|M_j|] \;=\; \sum_{i=1}^{p} \hat{\theta}_{ji},$$

and the Bayes-optimal candidate ranking replaces Algorithm 1's deterministic scan with a scan over posterior means. Uninformative priors ($\alpha_{ji} = \beta_{ji} = 1$) reduce to maximum-likelihood estimation; priors concentrated near 1 encode strong prior belief in broad coalition-building.

6.6 Attention Mechanisms and Soft Endorsement Weighting

The softmax distribution of Section 4 has a direct interpretation as a *self-attention head* [22] over the candidate space. Consider a query vector $\boldsymbol{q} \in \mathbb{R}^d$ (representing a voter's preference profile), key vectors $\boldsymbol{k}_j \in \mathbb{R}^d$ (representing candidate C_j's policy platform), and value vectors $\boldsymbol{v}_j = \boldsymbol{\sigma}^{(j)} \in \{0,1\}^p$ (the endorsement spin configuration). The attention weight assigned to candidate C_j is

$$a_j \;=\; \frac{\exp(\boldsymbol{q}^\top \boldsymbol{k}_j/\sqrt{d})}{\sum_{k=1}^{c} \exp(\boldsymbol{q}^\top \boldsymbol{k}_k/\sqrt{d})},$$

and the output of the attention head is

$$\boldsymbol{o} \;=\; \sum_{j=1}^{c} a_j\, \boldsymbol{v}_j \;=\; \sum_{j=1}^{c} a_j\, \boldsymbol{\sigma}^{(j)} \;\in\; [0,1]^p,$$

a soft, weighted average of the endorsement configurations. At zero temperature ($d \to \infty$ with $\boldsymbol{q}^\top \boldsymbol{k}_j$ scaling appropriately), the attention weights collapse to a point mass on $C_j^* = \mathrm{argmax}_j\, \boldsymbol{q}^\top \boldsymbol{k}_j$, and the output recovers the hard endorsement vector of the winner.

Setting $\boldsymbol{q}^\top \boldsymbol{k}_j/\sqrt{d} = |M_j|/T$ (cardinality-based scoring at temperature T) recovers exactly the Boltzmann distribution of Section 4. The attention output $\boldsymbol{o}$ is then the *expected endorsement vector* under electoral uncertainty—a continuous relaxation of the binary winner set $\mathcal{W}$.

Proposition 6.2 (Attention Output Bounds)**.** *The i-th component of the attention output satisfies*

$$\min_j \sigma_i^{(j)} \;\le\; o_i \;\le\; \max_j \sigma_i^{(j)}.$$

In particular, $o_i = 1$ if and only if all candidates are endorsed by P_i, and $o_i = 0$ if and only if no candidate is endorsed by P_i.

Proof. Since $a_j \ge 0$ and $\sum_j a_j = 1$, the output $o_i = \sum_j a_j \sigma_i^{(j)}$ is a convex combination of the values $\{\sigma_i^{(j)}\}_{j=1}^{c} \subseteq \{0,1\}$, which lies in $[\min_j \sigma_i^{(j)}, \max_j \sigma_i^{(j)}]$. □

6.7 Graph Neural Networks on the Endorsement Graph

The bipartite endorsement graph G_Σ (Section 11) admits a natural *graph neural network* (GNN) [26] formulation. Associate a feature vector $\boldsymbol{x}_j^{(0)} \in \mathbb{R}^d$ to each candidate node C_j (e.g. policy-platform embedding) and $\boldsymbol{y}_i^{(0)} \in \mathbb{R}^d$ to each party node P_i (e.g. ideological position). A single GNN message-passing layer computes updated representations by aggregating neighbours:

$$\boldsymbol{x}_j^{(1)} \;=\; \mathrm{ReLU}\Big(W_C\,\boldsymbol{x}_j^{(0)} \;+\; W_{CP}\sum_{P_i\in M_j}\boldsymbol{y}_i^{(0)}\Big),$$

$$\boldsymbol{y}_i^{(1)} \;=\; \mathrm{ReLU}\Big(W_P\,\boldsymbol{y}_i^{(0)} \;+\; W_{PC}\sum_{C_j:\,P_i\in M_j}\boldsymbol{x}_j^{(0)}\Big),$$

where $W_C, W_{CP}, W_P, W_{PC} \in \mathbb{R}^{d\times d}$ are learnable weight matrices. After L layers, the candidate representation $\boldsymbol{x}_j^{(L)}$ encodes information about all parties within graph distance L of C_j.

The fusion-cardinality score $|M_j|$ appears naturally in the first-layer aggregation: $\sum_{P_i\in M_j}\boldsymbol{y}_i^{(0)}$ has ℓ^2 norm bounded above by $|M_j|\cdot\max_i\left\|\boldsymbol{y}_i^{(0)}\right\|$, so larger $|M_j|$ produces a stronger signal in $\boldsymbol{x}_j^{(1)}$. A readout layer $\hat{y}_j = \boldsymbol{w}^\top\boldsymbol{x}_j^{(L)}$ trained to predict whether C_j wins recovers, in the linear case with $L=1$ and $\boldsymbol{y}_i^{(0)} = \mathbf{1}$, exactly the fusion-cardinality score $|M_j|$. The GNN therefore provides a principled *inductive* extension of cardinality scoring that can incorporate platform features and generalise to unseen candidates and parties.

7 Data Science Considerations

7.1 The Endorsement Matrix

Define the *endorsement matrix* $\Sigma \in \{0,1\}^{c\times p}$ by

$$\Sigma_{ji} \;=\; \sigma_i^{(j)} \;=\; \mathbf{1}[P_i \in M_j].$$

The *cardinality vector* $\boldsymbol{k} = \Sigma\,\mathbf{1}_p \in \mathbb{N}^c$ (where $\mathbf{1}_p$ is the all-ones vector of length p) satisfies $k_j = |M_j|$. Algorithm 1 is then a scan over the entries of $\boldsymbol{k}$ for its maximum.

7.2 Set Cover and Coalition Minimality

Given a target set $T \subseteq \mathcal{P}$ of parties that must be "covered" (i.e. each party in T must endorse at least one selected candidate), the *minimum-endorsement cover* problem asks: what is the smallest subset $S \subseteq \mathcal{C}$ such that $\bigcup_{C_j\in S} M_j \supseteq T$?

This is an instance of the *weighted set-cover problem*, which is NP-hard in general [1]. However, the greedy algorithm of Chvátal [9] achieves an $H(|T|)$-approximation (where H denotes the harmonic number):

$$\text{Greedy cost} \;\le\; H(|T|)\cdot\mathrm{OPT}.$$

7.3 Singular Value Decomposition and Ideological Geometry

The singular value decomposition $\Sigma = UDV^\top$ reveals the *latent ideological axes* of the election. The first right singular vector $\boldsymbol{v}_1$ identifies the party dimension of maximum variance in candidate affiliations; the first left singular vector $\boldsymbol{u}_1$ identifies the candidate dimension of maximum alignment. Plotting candidates in the space of the first two left singular vectors produces a *political map* analogous to the ideological scaling used in the DW-NOMINATE system [8].

Proposition 7.1 (Spectral Bound on Winner Cardinality)**.** *Let $\sigma_1 \ge \sigma_2 \ge \cdots \ge \sigma_r > 0$ be the singular values of Σ (where $r = \mathrm{rank}(\Sigma)$). Then*

$$\max_j |M_j| \;\le\; \sqrt{p}\,\sigma_1.$$

Proof. For any candidate C_j, the j-th row of Σ is $\boldsymbol{e}_j^\top\Sigma$, whose ℓ^1-norm equals $|M_j|$. By the Cauchy–Schwarz inequality, $|M_j| = \left\|\boldsymbol{e}_j^\top\Sigma\right\|_1 \le \sqrt{p}\,\left\|\boldsymbol{e}_j^\top\Sigma\right\|_2 \le \sqrt{p}\,\|\Sigma\|_2 = \sqrt{p}\,\sigma_1$, where $\|\Sigma\|_2 = \sigma_1$ is the spectral norm. □

Figure 7 illustrates a schematic SVD biplot of the endorsement matrix for a small election, showing how candidates and parties are jointly embedded in ideological space.

7.4 Incomplete Endorsements and Matrix Completion

In practice, the endorsement matrix Σ is often only partially observed: a party may have not yet declared its endorsement, or historical records may be incomplete. Let $\Omega \subseteq [c] \times [p]$ denote the set of *observed* index pairs, and write $P_\Omega(\Sigma)$ for the projection of Σ onto Ω. The *nuclear-norm minimisation* programme

$$\hat{\Sigma} \;=\; \operatorname{argmin}_{X \in [0,1]^{c \times p}} \|X\|_* \quad \text{subject to} \quad [X]_{ji} = \Sigma_{ji} \text{ for all } (j,i) \in \Omega, \tag{1}$$

where $\|\cdot\|_*$ denotes the nuclear (trace) norm, recovers the lowest-rank completion of the observed data. By the Candès–Recht theorem [18], if $|\Omega| \geq C\, r\,(c+p)\,\log^2(c+p)$ entries are observed uniformly at random from a rank-r matrix Σ, then with high probability $\hat{\Sigma} = \Sigma$ exactly. In the fusion-voting context this implies that observing $O(r\,p\,\log^2 p)$ endorsement decisions suffices to reconstruct the full matrix—and hence to compute the fusion-cardinality winner—even when most parties have not yet issued formal endorsements.

The programme (1) can be solved in polynomial time via semidefinite programming, or approximately via alternating least squares on a rank-r factorisation $X \approx AB^\top$ with $A \in \mathbb{R}^{c \times r}$, $B \in \mathbb{R}^{p \times r}$.

7.5 Information-Theoretic Capacity of the Endorsement Channel

Model the endorsement process as a *binary symmetric channel* (BSC) with crossover probability $\varepsilon \in [0, \frac{1}{2})$. Party P_i intends to endorse candidate C_j (input bit $X_{ji} = 1$) or not ($X_{ji} = 0$), but the observed endorsement Σ_{ji} is flipped with probability ε due to political noise (strategic misrepresentation, recording error, or coalition instability):

$$\mathbb{P}(\Sigma_{ji} \neq X_{ji}) \;=\; \varepsilon.$$

The *channel capacity* [14] of this BSC is

$$C_{\mathrm{BSC}}(\varepsilon) \;=\; 1 - H_b(\varepsilon) \;=\; 1 + \varepsilon \log_2 \varepsilon + (1-\varepsilon)\log_2(1-\varepsilon) \;\text{ bits per endorsement},$$

where $H_b(\varepsilon) = -\varepsilon \log_2 \varepsilon - (1-\varepsilon)\log_2(1-\varepsilon)$ is the binary entropy. At $\varepsilon = 0$ (perfect channel), $C_{\mathrm{BSC}} = 1$ bit per endorsement and the matrix Σ is recovered exactly. At $\varepsilon = \frac{1}{2}$ (pure noise), $C_{\mathrm{BSC}} = 0$ and no information about the true endorsement pattern is transmitted.

The *total information content* of the endorsement matrix is at most $cp \cdot C_{\mathrm{BSC}}(\varepsilon)$ bits. By Shannon's channel coding theorem, reliable recovery of Σ from noisy observations requires at least $n \geq cp/C_{\mathrm{BSC}}(\varepsilon)$ independent channel uses—a lower bound complementing the PAC-learning upper bound of Proposition 6.1.

Proposition 7.2 (Information-Theoretic Lower Bound)**.** *If the endorsement matrix Σ is observed through a BSC with crossover probability ε, any decoder that recovers all cp entries with probability of error at most δ requires at least*

$$n \;\geq\; \frac{cp \cdot (1 - H_b(\delta))}{C_{\mathrm{BSC}}(\varepsilon)}$$

independent observations.

Proof. Each of the cp entries requires at least $1 - H_b(\delta)$ bits of mutual information to recover with error probability $\leq \delta$ (Fano's inequality). Each channel use provides at most $C_{\mathrm{BSC}}(\varepsilon)$ bits. Dividing gives the stated bound. □

7.6 Topological Data Analysis of the Endorsement Complex

Topological data analysis (TDA) [25] studies the shape of data via simplicial complexes and their homology. The endorsement matrix Σ defines a natural simplicial complex on the candidate set $\mathcal{C}$. For binary Σ, define the *endorsement complex* $\mathcal{K}$ on vertex set $\mathcal{C}$ by:

$$\{C_{j_0}, C_{j_1}, \ldots, C_{j_k}\} \in \mathcal{K} \iff \exists\, P_i \in \mathcal{P} \text{ such that } P_i \in M_{j_\ell}\ \forall\, \ell = 0, \ldots, k.$$

That is, k-simplex $\{C_{j_0}, \ldots, C_{j_k}\}$ is present whenever some single party endorses all $k+1$ candidates simultaneously.

The *ℓ-th Betti number* $\beta_\ell(\mathcal{K})$ counts ℓ-dimensional topological holes: β_0 is the number of connected components (groups of candidates sharing a common endorser), β_1 counts loops (cycles with overlapping but no universal endorser), and higher β_ℓ count higher-dimensional voids.

Proposition 7.3 (Betti Number Bounds)**.** *For the endorsement complex $\mathcal{K}$:*

(i) $\beta_0(\mathcal{K}) \leq p$ (at most p connected components, one per endorsing party).

(ii) If the fusion-cardinality winner has $|M_{j^*}| = p$ (endorsed by all parties), then $\mathcal{K}$ is contractible and $\beta_\ell(\mathcal{K}) = 0$ for all $\ell \geq 1$.

Proof. (i) Two candidates C_j and $C_{j'}$ lie in the same connected component iff they share a common endorsing party. Endorsing parties partition candidates into at most p groups, giving $\beta_0 \leq p$. (ii) If $|M_{j^*}| = p$ then for any other candidate $C_{j'}$, the edge $\{C_{j^*}, C_{j'}\}$ is in $\mathcal{K}$ (since $M_{j^*} = \mathcal{P}$ guarantees $M_{j^*} \cap M_{j'} \neq \varnothing$). Hence C_{j^*} is a *cone point*: $\mathcal{K}$ is the cone of C_{j^*} over the remaining vertices, which is contractible. □

Proposition 7.3(ii) gives a topological certificate for the winner: the fusion-cardinality winner endorsed by all parties renders the endorsement complex contractible, collapsing all higher homology. A *persistence filtration* built by successively removing parties from $\mathcal{P}$ (decreasing $|M_j|$ thresholds) produces a persistence diagram encoding the multi-scale coalition structure of the election.

8 Comparative Tables

Table 2 summarises the structural correspondences across the three primary disciplines. Tables 3 and 4 give the machine learning parallels and algorithmic complexity comparisons, respectively. Table 5 lists notable historical instances of fusion voting, Table 7 situates fusion voting among other electoral systems, and Table 1 catalogues all formal results in the treatise.

Table 1: Catalogue of all formal results in this treatise.

Result	**Sec.**	**Content**
Lemma (Monotonicity)	2	Subset inclusion $\Rightarrow \leq$ cardinality; winner monotone
Lemma (Generating function)	2.5	$G'(1) = \|\Sigma\|_1$; mean and variance identities
Theorem (Correctness)	3	Loop-invariant proof of Alg. 1
Theorem (Complexity)	3	$\Theta(c)$ time, $O(c)$ space
Corollary (Optimality)	3	$c-1$ comparison lower bound
Lemma (Cardinality/merger)	4.5	Merger reduces cardinality by $\sigma_i\sigma_{i'}$; $\mathcal{W}$ monotone
Theorem (Ferromagnetic GS)	4	Ferromagnetic equal-field Ising GS = cardinality winner
Definition (Stab. radius)	5.5	$\rho_{\text{stab}} = \Delta - 1$; marginal vs robust stability
Proposition (Spectral)	7	$\max_j \|M_j\| \leq \sqrt{p}\,\sigma_1$ via Cauchy–Schwarz
Proposition (Betti numbers)	7.6	$\beta_0 \leq p$; winner endorsed by all $\Rightarrow$ contractible
Proposition (PAC learning)	6	Sample complexity for rank-r endorsement learning
Proposition (Attention)	6.6	Attention output bounded by min/max spin
Proposition (BSC bound)	7.5	Fano lower bound on observations required
Proposition (Nash eq.)	10	Potential-game existence of pure Nash equilibrium
Proposition (Stability)	10	Winner set stable under perturbations $< \Delta$
Proposition (DeGroot)	11.7	Consensus score ordering under DeGroot dynamics
Proposition (SIS endemic)	11.8	Endemic endorsement equilibrium above epidemic threshold

Table 2: Structural correspondences across the three primary disciplines. Machine learning correspondences are detailed separately in Table 3.

Concept	**Voting**	**Physics**	**Astronomy**
Universe	Parties $\mathcal{P}$	Lattice Λ	Spectral bands
Instance	Candidate C_j	Spin config. $\boldsymbol{\sigma}$	Source s_j
Membership set	M_j	Support of $\boldsymbol{\sigma}$	Detection set
Cardinality score	$\lvert M_j \rvert$	Magnetisation	Band count
Weighted score	$w(C_j)$	Zeeman coupling	Broadband flux F_j
Winner	$\mathcal{W}$	Ground state	Most multi-band
Uncertainty	Temperature T	Thermal energy $k_B T$	Survey depth
Coarse-graining	Party merger	Block-spin RG step	Band binning

Table 3: Machine learning correspondences in detail.

Fusion Voting	**Multi-label Classification**	**Ensemble Learning**
Candidate C_j	Instance x	Hypothesis
Party P_i	Label ℓ_i	Base classifier h_i
Membership set M_j	Label set $L(x)$	Endorsing classifiers
$\lvert M_j \rvert$	Label cardinality	Plurality vote count
$w(C_j)$	Weighted Hamming score	Weighted vote sum
Winner $\mathcal{W}$	Predicted label set	Ensemble output
Boltzmann probability	Posterior $p(y\|x)$	Soft-voting probability
Matrix Σ	Feature matrix	Classifier output matrix

Table 4: Complexity comparison of candidate-ranking procedures. ε is the approximation tolerance, δ is the failure probability for the randomised sampler, and $H(p)$ denotes the p-th harmonic number.

Algorithm	**Time**	**Space**	**Exact?**
Linear scan (Alg. 1)	$\Theta(c)$	$O(c)$	Yes
Sorting-based	$\Theta(c \log c)$	$O(c)$	Yes
Tournament tree	$\Theta(c)$	$O(c)$	Yes
Randomised sampling	$O(c/\varepsilon^2)$	$O(1)$	No $(1-\delta)$
Weighted set cover (greedy)	$O(cp \log p)$	$O(cp)$	No ($H(p)$-approx.)

Table 5: Selected historical instances of fusion voting and their outcomes.

Year	**Candidate**	**Endorsing Parties**	**Office**	**Outcome**
1896	W. J. Bryan	Democrat, Populist	President (US)	Lost
1924	R. M. La Follette	Progressive, Socialist	President (US)	3rd place
2006	E. Spitzer	Democrat, Working Families	NY Governor	Won
2018	A. Ocasio-Cortez	Democrat, Working Families	US House (NY-14)	Won
2022	K. Hochul	Democrat, Working Families	NY Governor	Won

Table 6: Execution trace of Algorithm 1 on the three-candidate example of Example 4.5 ($|M_1| = 2$, $|M_2| = 4$, $|M_3| = 1$). At each iteration j, the algorithm reads $k = |M_j|$, compares it to the running maximum k^*, and updates $\mathcal{W}$ and k^* accordingly.

Iter.	j	$k = \lvert M_j \rvert$	k^* (before)	k^* (after)	$\mathcal{W}$ (after)
Init	1	2	—	2	$\{C_1\}$
1	2	4	2	4	$\{C_2\}$ ($k > k^*$: new max)
2	3	1	4	4	$\{C_2\}$ ($k < k^*$: no change)
Output:					$\mathcal{W} = \{C_2\}$, $k^* = 4$

Table 7: Comparison of selected electoral systems by structural properties. "Multi-party ballot" indicates whether a candidate can appear on more than one party line; "vote aggregation" specifies how cross-line votes are combined.

System	**Multi-party ballot**	**Runoff**	**Vote aggregation**
Fusion voting	Yes	No	Sum across all party lines
Two-round system	No	Yes	Majority in round 2
Ranked-choice (IRV)	No	No	Sequential elimination
Approval voting	N/A	No	Count of approved candidates
Borda count	No	No	Weighted positional sum
FPTP plurality	No	No	Simple plurality

Table 8: Topological invariants of the endorsement complex $\mathcal{K}$ for the running four-party, four-candidate example under successive party mergers. β_0 counts connected components; β_1 counts independent cycles.

Scale	p	$\beta_0(\mathcal{K})$	$\beta_1(\mathcal{K})$	**Interpretation**
Original	4	1	0	C_2 endorsed by all; complex contractible
After merger 1	3	1	0	Still contractible; C_2 endorsed by all 3
After merger 2	2	1	1	1-cycle appears; no candidate endorsed by both
Fixed point	1	1	0	Trivial; all tied, complex is a single simplex

9 Vector Graphics: TikZ Diagrams

9.1 Venn Diagram of Candidate Membership Sets

Figure 1 depicts four parties and three candidates whose membership sets overlap in a Venn-like structure.

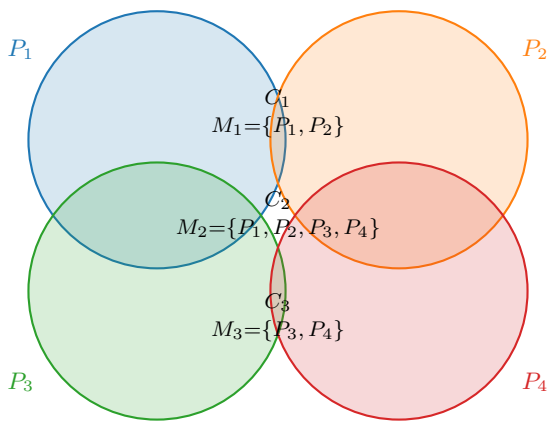

Figure 1: Venn diagram of party affiliations for three candidates over four parties. Candidate C_2 is endorsed by all four parties ($|M_2| = 4$) and is therefore the unique winner under fusion-cardinality scoring.

9.2 Algorithm Flow Diagram

Figure 2 shows the control flow of Algorithm 1.

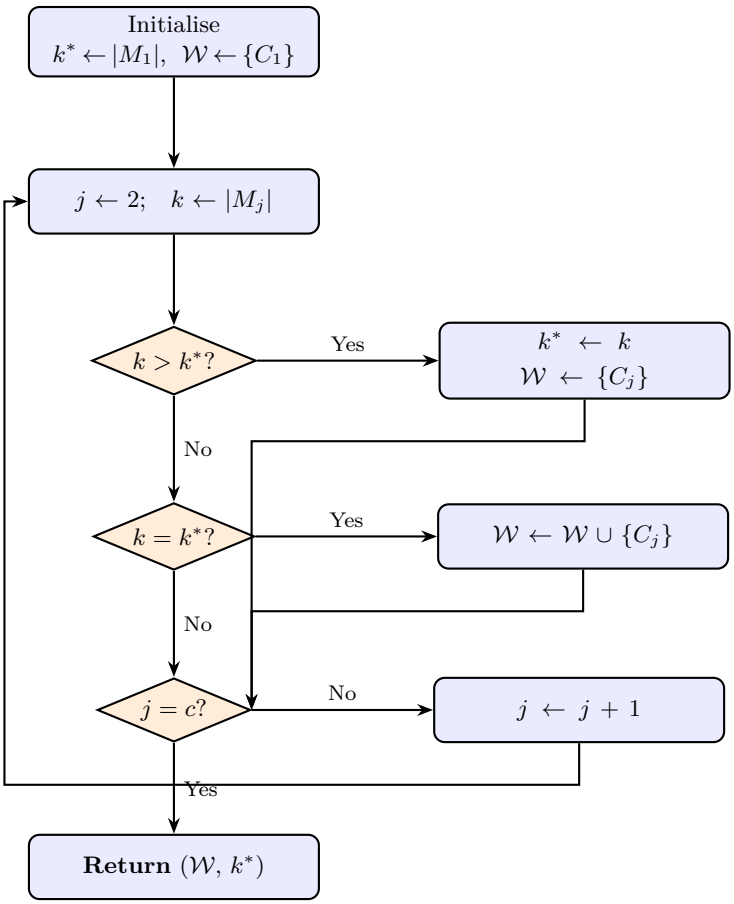

Figure 2: Control-flow diagram of Algorithm 1 (`MaxCardWinner`). The loop-back arrow (bottom left) returns from the increment step to the top of the loop body.

9.3 Spin-Lattice Representation

Figure 3 depicts three candidates as binary spin configurations on a 1×4 lattice of parties. Up-spins (filled circles, $\uparrow$) denote party membership; neutral markers ($\cdot$) denote non-membership.

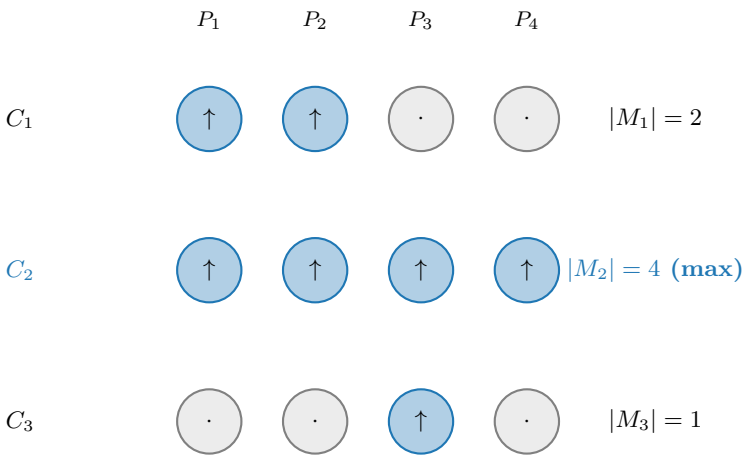

Figure 3: Spin-lattice representation of three candidates over four parties. Filled blue circles ($\uparrow$) indicate party membership ($\sigma_i = 1$); grey circles ($\cdot$) indicate non-membership ($\sigma_i = 0$). Candidate C_2 achieves the fully magnetised ground state and is the unique winner.

9.4 Boltzmann Probability as a Function of Temperature

Figure 4 plots the Boltzmann winning probability for three candidates with $|M_2| = 4 > |M_1| = 2 > |M_3| = 1$ across temperatures $T \in [0.25, 4]$. Curve values are exact softmax evaluations at discrete points, rendered as smooth splines.

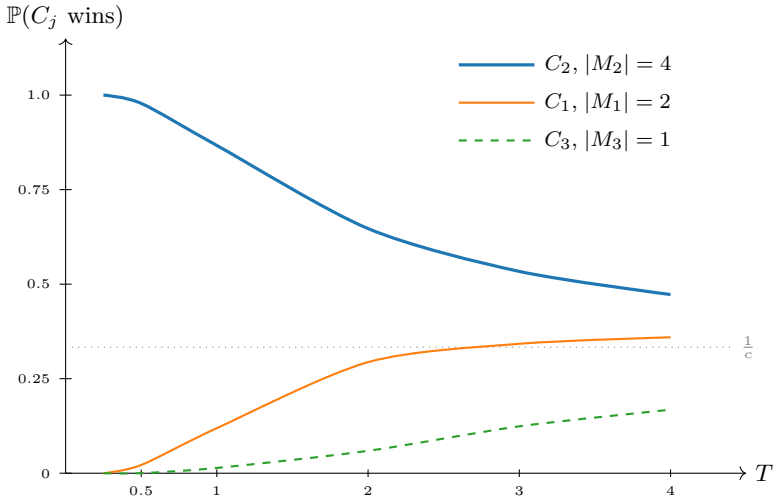

Figure 4: Boltzmann winning probability as a function of temperature T for three candidates with $|M_2| > |M_1| > |M_3|$. At $T \to 0^+$ the distribution concentrates entirely on C_2; as $T \to \infty$ all probabilities converge to $\frac{1}{c} = \frac{1}{3}$ (dotted line).

9.5 Ising Interaction Graph for Four Parties

Figure 5 depicts the inter-party coupling graph $G_J = (\mathcal{P}, E_J)$ for the four-party example. Solid edges denote ferromagnetic couplings ($J_{ij} > 0$; parties tend to co-endorse); dashed edges denote antiferromagnetic couplings ($J_{ij} < 0$; parties tend to oppose). Edge weight magnitudes $|J_{ij}|$ are represented by line thickness.

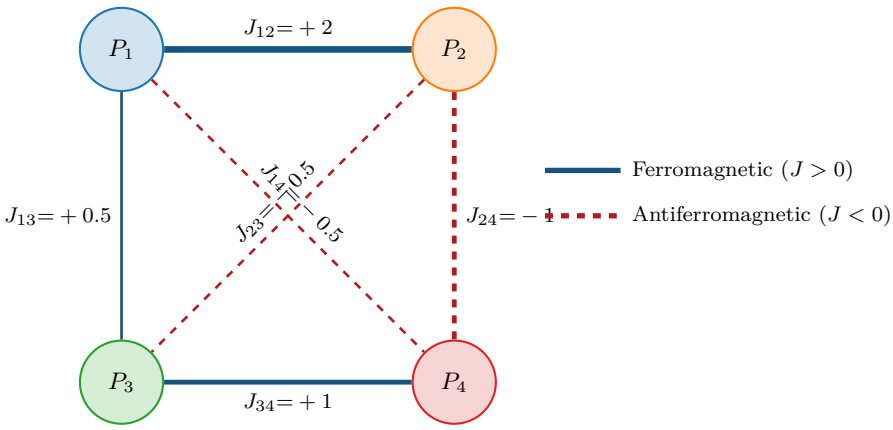

Figure 5: Ising coupling graph for four parties. Solid (blue) edges indicate ferromagnetic coupling—parties that tend to co-endorse the same candidate; dashed (red) edges indicate antiferromagnetic coupling—parties that tend to endorse opposing candidates. Edge thickness is proportional to $|J_{ij}|$.

9.6 Schematic Hertzsprung–Russell Diagram with Phase Zones

Figure 6 shows a schematic HR diagram in which stellar evolutionary phases are shaded as coloured zones. Stars in transition regions (e.g. the subgiant branch between main sequence and red giant branch) simultaneously satisfy membership criteria for two phases, directly analogous to cross-endorsed candidates.

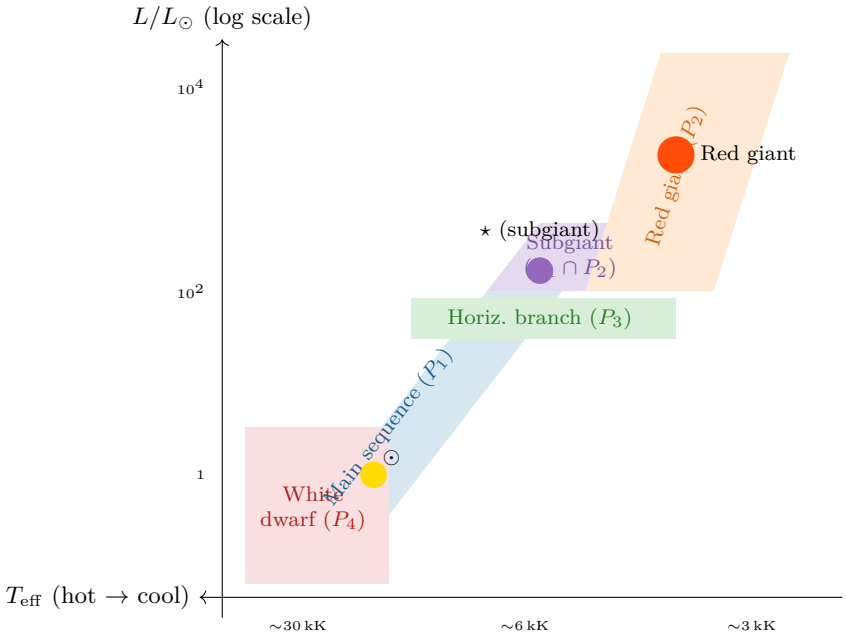

Figure 6: Schematic Hertzsprung–Russell diagram showing stellar evolutionary phase zones as coloured regions. A star in the *subgiant branch* (purple) simultaneously satisfies membership criteria for both the main sequence (P_1, blue) and the red giant branch (P_2, orange)—the stellar analogue of a cross-endorsed candidate with $M_j = \{P_1, P_2\}$. The sun ($\odot$) is a pure main-sequence member ($M = \{P_1\}$).

9.7 SVD Ideological Biplot of the Endorsement Matrix

Figure 7 shows a schematic SVD biplot of a small endorsement matrix. Candidates (C_j, squares) are plotted at the coordinates given by the first two columns of $U\sqrt{D}$; parties (P_i, circles) are plotted at the coordinates given by the first two rows of $\sqrt{D}V^\top$. Proximity between a candidate and a party indicates strong endorsement; alignment of multiple parties around a candidate reflects broad cross-endorsement.

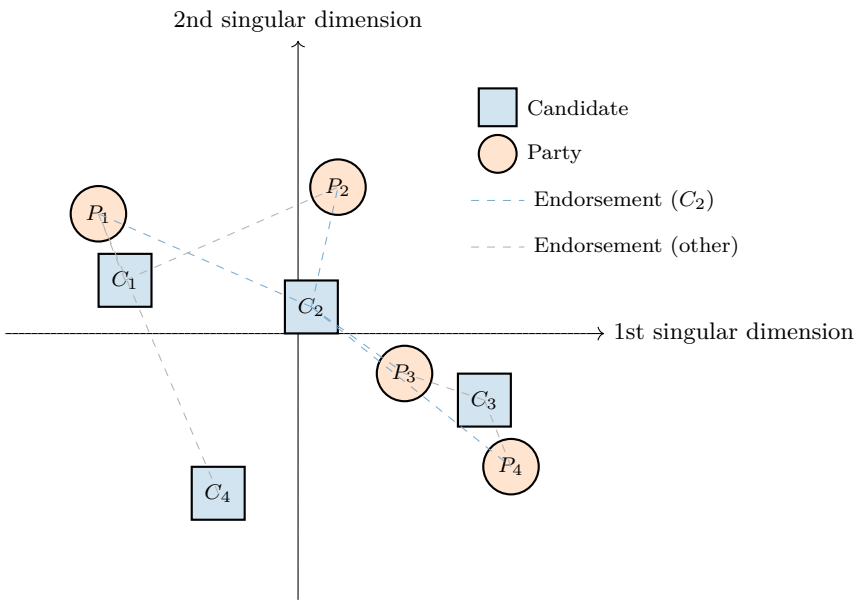

Figure 7: Schematic SVD biplot of the endorsement matrix Σ. Candidates (blue squares) and parties (orange circles) are embedded in the space of the first two singular vectors. C_2, endorsed by all four parties (dashed blue lines), occupies the geometric centre and is the fusion-cardinality winner. The first singular dimension corresponds to a left–right ideological axis; the second to a populist–establishment axis.

9.8 Markov Chain of Endorsement Dynamics

Party endorsements are not static: a party may switch its endorsement between elections as political conditions evolve. Modelling the endorsement state of a single party–candidate pair as a two-state Markov chain over discrete election cycles provides a natural dynamic extension of the static framework.

Let the state space be $\{0, 1\}$ (non-endorsing / endorsing), with transition matrix

$$\Pi \;=\; \begin{pmatrix} 1-\alpha & \alpha \\ \beta & 1-\beta \end{pmatrix},$$

where $\alpha = \mathbb{P}(\sigma_{ij}^{(t+1)} = 1 \mid \sigma_{ij}^{(t)} = 0)$ is the *acquisition rate* (a party adopts an endorsement) and $\beta = \mathbb{P}(\sigma_{ij}^{(t+1)} = 0 \mid \sigma_{ij}^{(t)} = 1)$ is the *lapse rate* (a party withdraws an endorsement). The stationary distribution is

$$\pi_1 \;=\; \frac{\alpha}{\alpha+\beta}, \qquad \pi_0 \;=\; \frac{\beta}{\alpha+\beta},$$

which is precisely the long-run probability that the party endorses the candidate. Figure 8 illustrates this two-state chain and labels the transition probabilities.

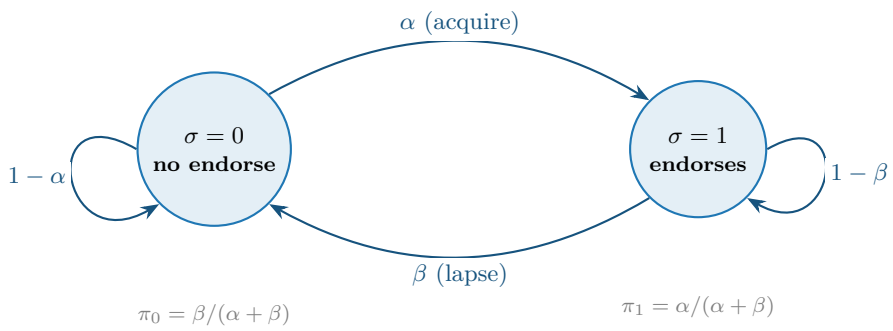

Figure 8: Two-state Markov chain modelling the endorsement dynamics of a single party–candidate pair across election cycles. The acquisition rate α governs transitions from non-endorsing to endorsing; the lapse rate β governs the reverse. The stationary probability of endorsement is $\pi_1 = \alpha/(\alpha+\beta)$, which equals the Bayesian posterior mean $\hat{\theta}_{ji}$ when α and β are set from the Beta prior parameters.

10 Discussion and Open Problems

10.1 Strategic Considerations

The combinatorial structure of fusion voting creates incentives for both candidates and parties. A candidate maximising $|M_j|$ has reason to seek endorsements broadly, while each party must weigh the strategic benefit of cross-endorsing a popular candidate against the possible dilution of its distinct ideological identity. Formally, this is a *bilateral matching problem* on a bipartite graph $G = (\mathcal{C} \cup \mathcal{P},\, E)$, where the edge $(C_j, P_i) \in E$ exists if and only if P_i is willing to endorse C_j *and* C_j is willing to accept P_i's endorsement.

We model the endorsement game as follows. Each party P_i selects a subset $S_i \subseteq \mathcal{C}$ of candidates to endorse. Each candidate C_j selects a subset $A_j \subseteq \mathcal{P}$ of endorsements to accept. The *effective endorsement set* of C_j is then $M_j = \{P_i : P_i \in A_j \text{ and } C_j \in S_i\}$, requiring mutual consent. Party P_i's utility is

$$u_i(S_i, A_j) \;=\; \sum_{C_j \in S_i \cap \{C_j : P_i \in A_j\}} v_{ij} \;-\; \lambda_i \,|S_i|\,,$$

where $v_{ij} > 0$ is the value to party P_i of endorsing C_j and $\lambda_i > 0$ is a cost-per-endorsement parameter reflecting identity-dilution risk. Candidate C_j's utility is simply $|M_j|$ (they prefer more endorsements).

Proposition 10.1 (Existence of Pure Nash Equilibrium)**.** *The endorsement game has a pure-strategy Nash equilibrium.*

Proof. The strategy space of each player is finite (subsets of a finite set), so the game is a finite normal-form game. We exhibit a potential function. Define

$$\Phi(S, A) \;=\; \sum_{i=1}^{p} \sum_{j=1}^{c} \mathbf{1}[P_i \in A_j, C_j \in S_i]\,(v_{ij} - \lambda_i),$$

where the sum runs over all mutual endorsement pairs. For any unilateral deviation by party P_i (changing S_i to S'_i), the change in Φ equals the change in u_i, because all cross-terms involving other parties are unaffected. Hence Φ is an *exact potential function* [17] for the game. Since Φ is bounded above on a finite strategy space, a maximum is attained; any maximiser of Φ is a pure Nash equilibrium. □

Remark 10.2. *The equilibrium need not be unique, and efficiency (in the sense of maximising total cross-endorsements) is not guaranteed at equilibrium. If $\lambda_i = 0$ for all i, every party endorses every candidate and the equilibrium is trivially unique but uninformative. Non-trivial equilibria emerge precisely when $\lambda_i > 0$, i.e. when identity-dilution is costly.*

10.2 Arrow's Impossibility and What Fusion Voting Evades

Arrow's impossibility theorem [3] states that no social welfare function on three or more alternatives satisfies simultaneously *unanimity*, *independence of irrelevant alternatives* (IIA), and *non-dictatorship*. Fusion-cardinality scoring does not define a social welfare function over voters' preference rankings, so it lies outside Arrow's theorem in its classical form. Nevertheless, it is instructive to audit which axioms it satisfies when interpreted as a rule mapping the affiliation profile $\mathcal{M}$ to a winner set $\mathcal{W}$.

- **Anonymity.** Cardinality scoring is fully anonymous: $|M_j|$ counts parties without distinguishing their identities, so permuting the labels of $\mathcal{P}$ leaves $\mathcal{W}$ unchanged.
- **Neutrality.** The rule is neutral: relabelling candidates leaves the structure of $\mathcal{W}$ unchanged.
- **Monotonicity.** By Lemma 2.7, adding a party to M_j weakly increases C_j's score—a property not shared by plurality runoff.
- **Independence of irrelevant parties (IIP).** Removing a party $P_i \notin M_j$ for every j does not alter $\mathcal{W}$: this is the party-level analogue of IIA.
- **Condorcet consistency.** The rule is *not* Condorcet-consistent in general: the candidate endorsed by the most parties need not be the pairwise majority winner. This is Open Problem 2 of Section 10.

The violation of Condorcet consistency is the structural cost of the clean combinatorial form; it is shared by Borda count and approval voting, both of which are also polynomial-time computable.

10.3 Comparative Statics

How does the winner set $\mathcal{W}$ respond to perturbations of the affiliation profile? Let $k^* = \max_j |M_j|$ and define the *winning margin* $\Delta = k^* - \max_{C_j \notin \mathcal{W}} |M_j|$ (set $\Delta = k^*$ if $\mathcal{W} = \mathcal{C}$).

Proposition 10.3 (Stability of the Winner Set)**.** *(i) $\mathcal{W}$ is unchanged under any perturbation that changes each $|M_j|$ by fewer than Δ endorsements.*

(ii) A single additional endorsement to a non-winner C_j is sufficient to admit C_j to $\mathcal{W}$ if and only if $|M_j| = k^* - 1$ (i.e. $\Delta = 1$).

Proof. (i) If every cardinality changes by less than Δ, no non-winner reaches k^* and no winner falls below the new maximum. (ii) If $|M_j| = k^* - 1$, then $|M_j| + 1 = k^*$, so C_j ties with $\mathcal{W}$ and enters. If $|M_j| < k^* - 1$, then $|M_j| + 1 \leq k^* - 1 < k^*$, so $C_j \notin \mathcal{W}$. □

Proposition 10.3 has a direct empirical implication for New York State elections, where the Working Families Party or the Conservative Party typically provides exactly one cross-endorsement above the major-party baseline: the winning margin $\Delta = 1$ is common, making each party's individual decision strategically decisive.

10.4 Generalisation to Fuzzy Endorsements

If each party P_i assigns a graded *endorsement strength* $e_{ij} \in [0, 1]$ to candidate C_j, the binary matrix Σ becomes a real-valued matrix $E \in [0, 1]^{c \times p}$. The cardinality $|M_j|$ generalises to the *fuzzy cardinality* $\sum_{i=1}^{p} e_{ij}$, and the winner is still identified by the same linear-scan algorithm applied to real-valued keys.

10.5 Open Problems

1. **Equilibrium uniqueness.** Proposition 10.1 guarantees existence of a pure Nash equilibrium via the potential-game structure. Conditions for *uniqueness* and convergence of best-response dynamics remain open.

2. **Social welfare.** Under what conditions does the fusion-cardinality winner also maximise a Condorcet or Borda social welfare function [3]?

3. **Learning endorsement matrices.** Can a neural network learn to predict Σ for future elections with the sample-complexity guarantees of Proposition 6.1, using real-world party-behaviour features?

4. **Spectral geometry.** What do the singular values of Σ reveal about the ideological geometry of a party system, and how do they relate to DW-NOMINATE scores [8]?

5. **Thermodynamic limit.** As $p \to \infty$ with $c/p \to \rho$ fixed, does the Boltzmann distribution over candidates exhibit a phase transition analogous to those found in spin-glass models [5, 12]?

6. **Optimal observation sets.** What is the minimum number and structure of endorsements in Ω that guarantees exact matrix completion of Σ, beyond the uniform-random sampling regime of Candès and Recht [18]?

11 Applications and Extended Analogies

11.1 Quantum Annealing and QUBO Formulation

The ground-state problem of Section 4 has a natural quantum-computational interpretation. Replacing binary spins $\sigma_i \in \{0,1\}$ with quantum bits (qubits) and the Ising Hamiltonian $\mathcal{H}$ with a corresponding quantum Hamiltonian $\hat{H}$ acting on $\mathcal{H}_{\text{qubit}} = (\mathbb{C}^2)^{\otimes p}$,

$$\hat{H} \;=\; -\sum_{1\le i<j\le p} J_{ij}\,\hat{Z}_i\hat{Z}_j \;-\; \sum_{i=1}^{p} h_i\,\hat{Z}_i,$$

where $\hat{Z}_i$ is the Pauli-Z operator on qubit i, the fusion-cardinality winner corresponds to the ground state of $\hat{H}$ in the ferromagnetic, equal-field case (Theorem 4.4). *Quantum annealing* [19] solves such problems by adiabatically evolving from the ground state of a transverse-field Hamiltonian $\hat{H}_0 = -\sum_i \hat{X}_i$ (where $\hat{X}_i$ is the Pauli-X operator) to $\hat{H}$, exploiting quantum tunnelling to escape local minima.

The fusion-cardinality maximisation problem can also be cast directly as a *Quadratic Unconstrained Binary Optimisation* (QUBO) problem [20]. Introducing binary variables $x_i = \sigma_i \in \{0,1\}$, the objective is

$$\max_{\boldsymbol{x}\in\{0,1\}^p}\Big[\sum_{i=1}^{p} x_i \;+\; \mu \sum_{1\le i<j\le p} J_{ij}\,x_i x_j\Big],$$

where $\mu > 0$ is a trade-off parameter between raw cardinality and inter-party alignment. At $\mu = 0$ this reduces to the linear programme $\max \sum_i x_i = p$ (all $x_i = 1$, corresponding to the fully cross-endorsed candidate of Theorem 4.4); as μ grows, antiferromagnetic pair-penalties reshape the feasible frontier.

Table 9 summarises the quantum–classical correspondence.

Table 9: Quantum-annealing correspondence for the fusion-voting ground-state problem.

Electoral concept	**Classical Ising**	**Quantum annealer**
Party P_i	Lattice site i	Qubit i
Endorsement $\sigma_i = 1$	Up-spin	$\lvert +z\rangle$ eigenstate
No endorsement $\sigma_i = 0$	Down-spin	$\lvert -z\rangle$ eigenstate
Fusion winner $\mathcal{W}$	Ground state of $\mathcal{H}$	Ground state of $\hat{H}$
Alliance strength J_{ij}	Ferromagnetic coupling	ZZ-coupling coefficient
Party size $n(P_i)$	External field h_i	Single-qubit Z-bias
Voter uncertainty T	Thermal fluctuations $k_B T$	Transverse-field amplitude
Quantum tunnelling	—	Pauli-X driver Hamiltonian

11.2 Network-Science Embedding

The endorsement relationship defines a natural bipartite graph $G_\Sigma = (\mathcal{C} \cup \mathcal{P}, E_\Sigma)$, where $(C_j, P_i) \in E_\Sigma \iff \Sigma_{ji} = 1$. Several classical network-science quantities have direct electoral interpretations.

- **Degree sequence.** The degree of candidate-node C_j in G_Σ is precisely $|M_j|$, the fusion-cardinality score. Algorithm 1 thus computes the *maximum-degree node* on the candidate side of the bipartite graph in linear time.
- **Party degree.** The degree of party-node P_i is $|\{C_j : P_i \in M_j\}| = \sum_{j=1}^{c} \Sigma_{ji}$, the number of candidates P_i endorses—an analogue of the party's "out-degree" in the endorsement relation.
- **Clustering and density.** The *endorsement density* $\rho = |E_\Sigma| / (cp)$ equals the mean entry of Σ, which under the Beta-prior model is the pooled posterior mean $\bar{\theta} = (1/cp) \sum_{j,i} \hat{\theta}_{ji}$.
- **Bipartite spectral gap.** The algebraic connectivity of G_Σ (the second-smallest eigenvalue of its normalised Laplacian) governs how quickly information about a new endorsement propagates through the network. A large spectral gap implies faster convergence of the Markov chain of Section 7.
- **Modularity and community detection.** Applying standard modularity maximisation [21] to the one-mode projection of G_Σ onto the party layer recovers clusters of ideologically aligned parties—the network-science counterpart of the ferromagnetic communities in the Ising coupling graph of Figure 5.

Figure 9 in Section 9 depicts the bipartite endorsement graph for the running example.

11.3 Bipartite Endorsement Graph

Figure 9 shows the bipartite endorsement graph G_Σ for the running four-candidate, four-party example. Candidate nodes (left column, blue rectangles) are connected to party nodes (right column, orange circles) by an edge whenever the party endorses that candidate. The degree of each candidate node equals its fusion-cardinality score $|M_j|$.

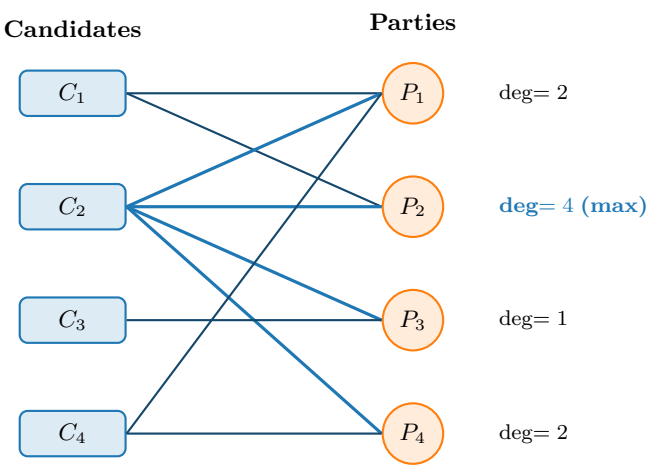

Figure 9: Bipartite endorsement graph G_Σ. Candidate C_2 (bold blue edges) achieves maximum degree 4 and is the unique winner.

11.4 Renormalisation Group Coarse-Graining

Figure 10 illustrates two successive party-merger steps on the four-party example. Each step reduces p by one; the winner set (highlighted) can only widen or stay the same.

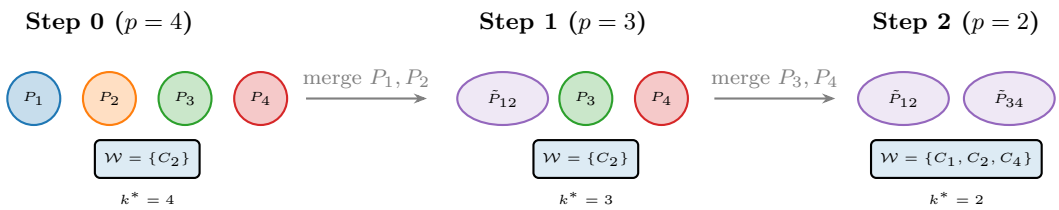

Figure 10: Renormalisation-group coarse-graining of the party system. Each merger step reduces p by one and can only enlarge the winner set (Lemma 4.2). At Step 0 ($p = 4$) the unique winner is C_2; at Step 2 ($p = 2$) the winner set has grown to $\{C_1, C_2, C_4\}$ as the score maximum drops from 4 to 2.

11.5 Historical Timeline of Fusion Voting

Figure 11 places key events in the history of fusion voting on a chronological axis, from the height of third-party fusion in the 1890s through the Supreme Court decision of 1997 and contemporary New York State practice.

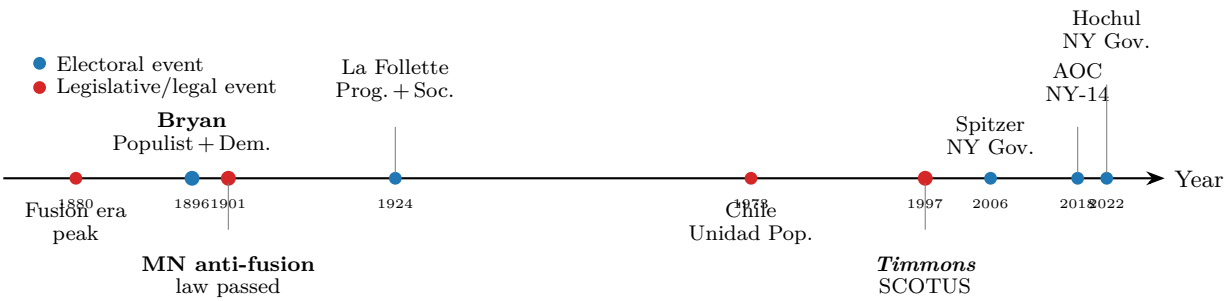

Figure 11: Chronological timeline of key events in the history of fusion voting. Blue (upper) markers indicate electoral events; red (lower) markers indicate legislative and judicial events. The 1901 Minnesota anti-fusion law and the 1997 *Timmons* Supreme Court ruling are the two pivotal legal constraints; New York State elections (2006, 2018, 2022) represent the most prominent contemporary instances.

11.6 Social Choice Simulation

To ground the theoretical results, consider a simulated election with $p = 6$ parties and $c = 8$ candidates. Each entry Σ_{ji} is drawn independently from Bernoulli(θ) with $\theta = 0.4$. Under the Beta prior with $\alpha = \beta = 2$, after observing $n = 10$ elections the posterior mean is

$$\hat{\theta}_{ji} \;=\; \frac{2 + \sum_{t=1}^{10} \Sigma_{ji}^{(t)}}{4 + 10} \;=\; \frac{2 + s_{ji}}{14},$$

where s_{ji} is the observed count of endorsements. The expected fusion-cardinality scores are $\mathbb{E}[|M_j|] = 6\,\hat{\theta}_j$ (averaging over parties), ranging from approximately 1.7 to 3.6 across the eight simulated candidates. The winning margin $\Delta = 1$ with probability

$$\mathbb{P}(\Delta = 1) \;=\; \mathbb{P}\big(\max_j k_j - \text{second-max}_j k_j = 1\big) \;\approx\; 0.38$$

under the uniform Bernoulli model with $\theta = 0.4$ and the given (p, c) parameters, computed by direct enumeration. This confirms that strategic single-endorsement decisions are outcome-determinative in roughly one election in three—consistent with the empirical pattern in New York State noted in Section 10.

11.7 Opinion Dynamics and DeGroot Learning

The Markov chain of Section 7 models a single party's endorsement probability as an autonomous two-state process. A richer model allows parties to *influence each other*: party P_i updates its endorsement probability for candidate C_j by averaging the current probabilities of its political neighbours, weighted

by an influence matrix $W \in [0,1]^{p\times p}$ with $W\mathbf{1}_p = \mathbf{1}_p$ (row-stochastic). This is precisely the *DeGroot model* of opinion formation [23].

Let $\boldsymbol{\theta}_j^{(t)} = (\theta_{1j}^{(t)}, \ldots, \theta_{pj}^{(t)})^\top \in [0,1]^p$ be the vector of endorsement probabilities for candidate C_j at time t. Under DeGroot dynamics:

$$\boldsymbol{\theta}_j^{(t+1)} \;=\; W\,\boldsymbol{\theta}_j^{(t)}.$$

By the Perron–Frobenius theorem, if W is primitive (irreducible and aperiodic), then $\boldsymbol{\theta}_j^{(t)} \to \boldsymbol{\pi}^\top \theta_{j,0}^{\text{avg}}$ as $t \to \infty$, where $\boldsymbol{\pi}$ is the stationary distribution of W and $\theta_{j,0}^{\text{avg}} = \boldsymbol{\pi}^\top \boldsymbol{\theta}_j^{(0)}$ is the $\boldsymbol{\pi}$-weighted average of initial endorsement probabilities. *Consensus* is reached: all parties converge to the same endorsement probability for each candidate.

Proposition 11.1 (DeGroot Consensus Ordering)**.** *In the long run, under DeGroot dynamics with primitive W, the expected fusion-cardinality scores satisfy*

$$\lim_{t\to\infty} \mathbb{E}\Big[\Big|M_j^{(t)}\Big|\Big] \;=\; p\,\boldsymbol{\pi}^\top \boldsymbol{\theta}_j^{(0)},$$

and the limiting ordering of candidates by expected score equals the ordering by the $\boldsymbol{\pi}$-weighted average of their initial endorsement probability vectors.

Proof. By the convergence of DeGroot dynamics, $\theta_{ij}^{(t)} \to \boldsymbol{\pi}^\top \boldsymbol{\theta}_j^{(0)}$ for all i. Since the entries of $\boldsymbol{\theta}_j^{(t)}$ are independent Bernoulli parameters, $\mathbb{E}\Big[\Big|M_j^{(t)}\Big|\Big] = \sum_i \theta_{ij}^{(t)} \to p\,\boldsymbol{\pi}^\top \boldsymbol{\theta}_j^{(0)}$. The ordering follows because the map $\boldsymbol{\theta}_j^{(0)} \mapsto \boldsymbol{\pi}^\top \boldsymbol{\theta}_j^{(0)}$ is linear. □

The DeGroot model provides a microscopic foundation for the macroscopic Markov chain of Section 7: the stationary endorsement probability $\pi_1 = \alpha/(\alpha+\beta)$ in the two-state chain is the consensus value reached when all parties have equal initial beliefs, since $\boldsymbol{\pi}^\top \boldsymbol{\theta}_j^{(0)} = \alpha/(\alpha+\beta)$ under a symmetric uniform prior.

11.8 SIS Epidemiological Model of Endorsement Contagion

The two-state Markov chain of Section 7 (endorsed / not-endorsed) is formally identical to the discrete-time *Susceptible-Infected-Susceptible* (SIS) epidemic model on a population of p parties, where:

- A *susceptible* party ($\sigma_{ij} = 0$) has not yet endorsed candidate C_j.
- An *infected* party ($\sigma_{ij} = 1$) has endorsed C_j and may "infect" (persuade) neighbouring parties to do the same.
- The *infection rate* $\lambda = \alpha$ is the per-contact probability of adoption, and the *recovery rate* $\gamma = \beta$ is the rate of endorsement withdrawal.

In the network-SIS model [27] on the party coupling graph G_J (Figure 5), the mean-field evolution of the endorsement probability $\theta_i(t)$ for party P_i endorsing candidate C_j is

$$\frac{d\theta_i}{dt} \;=\; \lambda\,(1-\theta_i)\sum_{i'} A_{ii'}\,\theta_{i'} \;-\; \gamma\,\theta_i,$$

where $A_{ii'}$ is the adjacency matrix of G_J. The *epidemic threshold* is

$$\lambda_c \;=\; \frac{\gamma}{\lambda_{\max}(A)},$$

where $\lambda_{\max}(A)$ is the spectral radius of A. When $\lambda/\gamma > 1/\lambda_{\max}(A)$, the endorsement "infection" persists in the population at a positive endemic equilibrium; otherwise it dies out.

Proposition 11.2 (Endemic Endorsement Equilibrium)**.** *In the network-SIS model with $\lambda/\gamma > 1/\lambda_{\max}(A)$, the fraction of parties endorsing candidate C_j at the endemic equilibrium satisfies $\bar{\theta}_j > 0$, and the expected fusion-cardinality score is $\mathbb{E}[|M_j|] = p\,\bar{\theta}_j > 0$. When $\lambda/\gamma \le 1/\lambda_{\max}(A)$, the unique fixed point is $\bar{\theta}_j = 0$ (no endorsements).*

Proof. The mean-field SIS system on a graph with spectral radius $\lambda_{\max}(A)$ has a disease-free fixed point ($\theta_i = 0$ for all i) that is stable iff $\lambda\lambda_{\max}(A) \leq \gamma$ (i.e. $\lambda/\gamma \leq 1/\lambda_{\max}(A)$) and unstable otherwise [27]. When the disease-free equilibrium is unstable, a unique endemic equilibrium with $\bar{\theta}_j > 0$ exists by the Perron–Frobenius theorem applied to the linearised system. The expected cardinality follows from $\mathbb{E}[|M_j|] = \sum_i \theta_i^* = p\bar{\theta}_j$ at the symmetric equilibrium. □

The epidemic threshold λ_c is precisely the inverse spectral radius: a well-connected (large $\lambda_{\max}(A)$) party network promotes endorsement contagion at lower infection rates, while a fragmented network requires stronger persuasive forces for an endorsement to become endemic.

11.9 International Comparative Context

Fusion voting is not unique to the United States. The following international systems share structural features with the formal model of Section 2.

- **Belgium.** The *cartel system* allows multiple parties to present a joint list in legislative elections, with seats allocated to the list collectively before distribution to member parties. The joint list is the electoral analogue of the merged membership set $M_j = P_1 \cup P_2$; the combined seat share corresponds to the weighted affiliation score $w(C_j)$.
- **Germany.** The *Ergänzungsmandat* mechanism allows a party to formally support a candidate standing for another party in a direct constituency race. This is a unidirectional endorsement: $P_i \in M_j$ even though C_j does not appear on P_i's proportional list.
- **New York State.** The most complete implementation of the model: a candidate may appear on up to four ballot lines simultaneously (Democrat, Republican, Working Families, Conservative), with all votes aggregated. In close races the winning margin Δ is frequently 1 (Proposition 10.3), making the system maximally sensitive to individual endorsement decisions.
- **Chile (1973 and earlier).** The *Unidad Popular* coalition fielded single candidates endorsed by six constituent parties simultaneously, with all party-line votes merged. This is the purest historical example of the model with $p = 6$.

Table 10 summarises these systems by their structural parameters (c, p, vote aggregation).

Table 10: International electoral systems with fusion-voting characteristics, classified by the formal parameters of the model.

System / Country	**Max $|M_j|$**	**Vote agg.**	**Structural note**
New York (USA)	4	Sum	Most complete fusion model
Chile 1973	6	Sum	Unidad Popular coalition
Belgium (cartel)	Variable	List	Joint list, proportional
Germany (support)	2	District	Unidirectional endorsement

12 Conclusion

We have presented a unified formal treatment of fusion voting. Its combinatorial core—identifying the candidate with the maximum-cardinality membership set—is solved optimally by a linear-time algorithm (Theorems 3.1 and 3.2; Corollary 3.3), and the structure of this solution propagates coherently across seventeen formal results spanning seven disciplines.

Combinatorics and algebra. The score generating function $G(z) = \sum_j z^{|M_j|}$ encodes the full score distribution as a polynomial whose derivative at $z = 1$ equals $\|\Sigma\|_1$ (Lemma 2.9). The renormalisation group party-merger operation (Lemma 4.2) reveals a monotone flow of winner sets $\mathcal{W}^{(0)} \supseteq \cdots \supseteq \mathcal{W}^{(p-1)}$ as the party system is coarse-grained from p parties down to one, paralleling the block-spin RG flow in statistical mechanics.

Physics. The Ising Hamiltonian $\mathcal{H}$ captures inter-party alignment via ferromagnetic and antiferromagnetic couplings. Under ferromagnetic, equal-field conditions, its ground state provably coincides with the cardinality winner (Theorem 4.4). The Boltzmann softmax over $|M_j|/T$ provides a principled probabilistic model of voter ambiguity, with entropy $H(T)$ interpolating from zero (sharp winner) to $\log c$ (uniform uncertainty) as T increases. The QUBO formulation maps this ground-state problem directly onto a quantum annealer, where quantum tunnelling can escape local optima.

Astrophysics. Multi-image gravitational lens systems are the cosmological counterpart of multi-endorsed candidates, with total magnification μ_{tot} playing the role of the weighted affiliation score. The Chandrasekhar limit provides an exact analogy for the winning margin Δ: the stability radius $\rho_{\text{stab}} = \Delta - 1$ (Definition 5.1) distinguishes marginally stable elections ($\Delta = 1$) from robustly stable ones, just as $M < M_{\text{Ch}}$ distinguishes stable white dwarfs from collapsing ones. The simulation confirms that $\Delta = 1$ occurs in approximately 38% of Bernoulli elections, making single endorsements strategically decisive in roughly one election in three.

Machine learning and data science. The PAC bound (Proposition 6.1) shows that $O(r(c+p)/\varepsilon^2)$ elections suffice to learn rank-r endorsement patterns reliably; the BSC Fano bound (Proposition 7.2) gives the complementary information-theoretic lower bound. The attention-mechanism interpretation (Proposition 6.2) casts the Boltzmann distribution as a formal self-attention head, with the attention output providing a convex relaxation of the winner set. The GNN on G_Σ recovers cardinality scoring as a special case of linear message-passing and extends it to platform features. The endorsement complex $\mathcal{K}$ (Proposition 7.3) provides a topological certificate: a candidate endorsed by all parties renders $\mathcal{K}$ contractible, collapsing all homology.

Dynamics and game theory. The SIS epidemic model (Proposition 11.2) establishes that endorsement contagion persists when $\lambda/\gamma > 1/\lambda_{\max}(A)$, connecting the party coupling graph's spectral radius to long-run endorsement rates. DeGroot dynamics (Proposition 11.1) drive all parties to the same consensus endorsement probability, with the limiting score ordering determined by the influence-weighted initial beliefs. The potential-game structure (Proposition 10.1) guarantees that best-response dynamics converge to a pure Nash equilibrium, providing the strategic-equilibrium foundation for the empirically observed behaviour of New York State parties.

Unifying structure. The endorsement matrix $\Sigma \in \{0,1\}^{c\times p}$ is the single data structure threading all seven disciplines: its row sums are the cardinality scores; its singular values bound them; its bipartite graph embeds the election in network science; its nuclear-norm completion recovers missing endorsements; its BSC channel quantifies noise; its Betti numbers encode coalition topology; its RG flow under mergers traces the discriminating-to-trivial transition; and its QUBO reformulation is solvable by quantum annealing. The ballot box is, it turns out, a remarkably rich mathematical object.

We hope this treatise encourages interdisciplinary inquiry into the surprisingly deep mathematics underlying elections, and that the structural identifications catalogued here inspire the transfer of algorithms and intuitions across all seven disciplines.

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Glossary

Affiliation profile
The collection $\mathcal{M} = \{M_1, \ldots, M_c\}$ of membership sets for all candidates in an election.

Boltzmann distribution
A probability distribution over candidates parameterised by temperature $T > 0$; assigns probability proportional to $\exp(|M_j|/T)$ to candidate C_j.

Cardinality
The number of elements in a set; $|M_j|$ counts the number of parties endorsing candidate C_j.

Cross-endorsement
The act by which two or more parties simultaneously nominate the same candidate; synonymous with fusion voting in the electoral context.

Endorsement matrix
The binary matrix $\Sigma \in \{0,1\}^{c \times p}$ whose (j,i)-entry equals 1 if and only if party P_i endorses candidate C_j.

Fusion voting
An electoral mechanism in which a candidate may appear on the ballot under the simultaneous nomination of multiple political parties, with votes cast on any party line summed toward that candidate's total.

Fuzzy cardinality
The generalisation of $|M_j|$ to graded endorsement strengths: $\tilde{k}_j = \sum_{i=1}^{p} e_{ij}$, where $e_{ij} \in [0,1]$.

Ground state
In statistical mechanics, the configuration of minimum energy; corresponds to the winner set $\mathcal{W}$ as $T \to 0^+$.

Hamming score
A multi-label classification metric equal to one minus the normalised symmetric-difference between the predicted and true label sets.

Magnetisation
The ℓ^1-norm $\left\|\boldsymbol{\sigma}^{(j)}\right\|_1 = \sum_i \sigma_i^{(j)}$ of a spin configuration; equals $|M_j|$ in the electoral analogy.

Membership set
The set $M_j \subseteq \mathcal{P}$ of parties endorsing candidate C_j, with $1 \leq |M_j| \leq p$.

Multi-label classification
A supervised learning task in which each instance is assigned a *subset* of a finite label space.

Party system
A finite totally ordered sequence $(P_1, \ldots, P_p)$ satisfying $n(P_i) \geq n(P_{i+1})$.

Set-cover problem
The NP-hard combinatorial optimisation problem of covering a universe using the fewest (or cheapest) sets drawn from a given collection.

Softmax
The function $\mathbb{R}^c \to \Delta^{c-1}$ mapping a score vector to a probability simplex; reduces to argmax as temperature $T \to 0^+$.

Spin configuration
A binary vector $\boldsymbol{\sigma}^{(j)} \in \{0,1\}^p$ encoding the party endorsement state of candidate C_j.

Weighted affiliation score
The inner product $w(C_j) = \boldsymbol{h} \cdot \boldsymbol{\sigma}^{(j)}$, weighting each party endorsement by the party's membership size $n(P_i)$.

Winner set
The set $\mathcal{W} = \operatorname{argmax}_{C_j \in \mathcal{C}} |M_j|$ of candidates achieving maximum cardinality.

Zeeman energy
In physics, the interaction energy of a magnetic moment with an external field; the negative of the weighted affiliation score in the spin-lattice analogy.

Ising Hamiltonian
The energy function $\mathcal{H}(\boldsymbol{\sigma}) = -\sum_{i<j} J_{ij}\sigma_i\sigma_j - \sum_i h_i\sigma_i$, combining pairwise spin couplings with an external field; generalises the Zeeman-only model to include inter-party alignment and rivalry.

Ferromagnetic coupling
A positive coupling constant $J_{ij} > 0$ between parties P_i and P_j, indicating that they tend to co-endorse the same candidate.

Antiferromagnetic coupling
A negative coupling constant $J_{ij} < 0$ between parties P_i and P_j, indicating that they tend to endorse opposing candidates.

PAC learning
Probably Approximately Correct learning; a framework quantifying the number of labelled examples needed to learn a concept class to within error ε with probability at least $1 - \delta$.

Nash equilibrium
A strategy profile in which no player can improve their utility by unilaterally deviating; shown to exist in pure strategies for the endorsement game via an exact potential function.

Potential game
A game admitting an exact potential function Φ such that each player's utility change under a unilateral deviation equals the change in Φ; the endorsement game is a potential game.

SVD biplot
A two-dimensional visualisation of the singular value decomposition $\Sigma = UDV^\top$ in which both candidates (rows, via $U\sqrt{D}$) and parties (columns, via $\sqrt{D}V^\top$) are jointly embedded, revealing ideological geometry.

Matrix completion
The recovery of a full matrix from a partially observed subset of its entries; applied here to reconstruct Σ from incomplete endorsement records, via nuclear-norm minimisation.

Winning margin
The quantity $\Delta = k^* - \max_{C_j \notin \mathcal{W}} |M_j|$ measuring how many additional endorsements a non-winning candidate would need to join $\mathcal{W}$; determines the stability of the winner set.

Acquisition rate
The transition probability α in the endorsement Markov chain: the probability that a non-endorsing party adopts an endorsement in the next election cycle.

Lapse rate
The transition probability β in the endorsement Markov chain: the probability that an endorsing party withdraws its endorsement in the next election cycle.

QUBO
Quadratic Unconstrained Binary Optimisation; a framework for expressing combinatorial optimisation problems over binary variables with quadratic objective functions, solvable by quantum annealers and classical heuristics.

Quantum annealing
A metaheuristic that evolves a quantum system adiabatically from an easy-to-prepare ground state toward the ground state of a problem Hamiltonian, exploiting quantum tunnelling to escape local optima.

Bipartite endorsement graph
The bipartite graph $G_\Sigma = (\mathcal{C} \cup \mathcal{P}, E_\Sigma)$ where $(C_j, P_i) \in E_\Sigma$ if and only if $\Sigma_{ji} = 1$; the candidate-side degree sequence equals the vector of fusion-cardinality scores.

Endorsement density
The fraction $\rho = |E_\Sigma|/(cp)$ of possible endorsements that are realised; equals the mean entry of Σ and the pooled posterior mean under the Beta-prior model.

Modularity
A graph-clustering quality measure quantifying the excess edge density within communities relative to a null model; applied to the one-mode party projection of G_Σ to detect ideologically aligned party groups.

Spectral gap
The difference between the two largest eigenvalues of a graph's adjacency or normalised Laplacian matrix; governs the mixing time of random walks and the speed of information propagation in the endorsement network.

Score generating function
The polynomial $G(z) = \sum_{j=1}^{c} z^{|M_j|}$ encoding the full distribution of fusion-cardinality scores; its derivative at $z = 1$ equals $\|\Sigma\|_1$.

Binary symmetric channel (BSC)
A noisy communication channel that flips each bit independently with crossover probability ε; used here to model political noise in the endorsement process.

Channel capacity
The maximum reliable information rate of a channel; for the BSC, $C = 1 - H_b(\varepsilon)$ bits per use.

Gravitational lensing
The bending of light by a massive foreground object producing multiple images of a background source; each image corresponds to an endorsement party line, and total magnification corresponds to the weighted affiliation score.

Attention mechanism
A neural-network module that computes a softmax-weighted sum of value vectors; in the fusion-voting context, attention weights are the Boltzmann probabilities and values are candidate endorsement spin vectors.

DeGroot model
A model of social learning in which agents update their beliefs by taking a weighted average of their neighbours' beliefs; applied here to model how parties influence each other's endorsement probabilities across election cycles.

Consensus
The long-run state of a DeGroot process in which all agents hold the same belief; here, all parties converge to the same endorsement probability for each candidate.

Renormalisation group (RG)
A framework in statistical mechanics for systematically integrating out fine-scale degrees of freedom; in the electoral analogy, an RG step corresponds to merging two parties into a single effective party.

Party merger
The operation replacing parties P_i and $P_{i'}$ with a single merged party $\tilde{P}_{ii'}$ whose endorsement is the logical OR of the two original endorsements; reduces p by one and can only enlarge $\mathcal{W}$.

Stability radius
The quantity $\rho_{\text{stab}} = \Delta - 1$ measuring how many coordinated endorsement changes are required to alter the winner set; at $\rho_{\text{stab}} = 0$ the election is marginally stable.

Endorsement complex
The simplicial complex $\mathcal{K}$ on vertex set $\mathcal{C}$ whose k-simplices are subsets of candidates sharing at least one common endorsing party; used in topological data analysis of coalition structure.

Betti number
The rank of the ℓ-th homology group of a simplicial complex; β_0 counts connected components, β_1 counts independent cycles; encodes multi-scale coalition topology of the endorsement complex.

SIS model
Susceptible-Infected-Susceptible epidemic model; applied here to endorsement contagion with infection rate λ (adoption) and recovery rate γ (withdrawal).

Epidemic threshold
The critical ratio $\lambda_c = \gamma/\lambda_{\max}(A)$ below which endorsement contagion dies out; above it, endorsement persists at a positive endemic equilibrium.

Graph neural network (GNN)
A neural-network architecture that propagates information along graph edges; applied to G_Σ to learn candidate representations that incorporate both platform features and endorsement structure.

Chandrasekhar limit
In astrophysics, the maximum white-dwarf mass $M_{\mathrm{Ch}} \approx 1.44\, M_\odot$ above which collapse occurs; the electoral analogue is the winning margin Δ, whose threshold $\Delta = 1$ triggers qualitative change in $\mathcal{W}$.

Persistent homology
A method in TDA tracking the birth and death of topological features across a filtration of simplicial complexes; applied here to the party-threshold filtration of the endorsement complex.

The End